QUT Centre for Robotics



ENN583 – Foundations of Robotic Vision

Week 7 – Sem 2 – 2026

Prof Niko Suenderhauf

Welcome to Week 7!

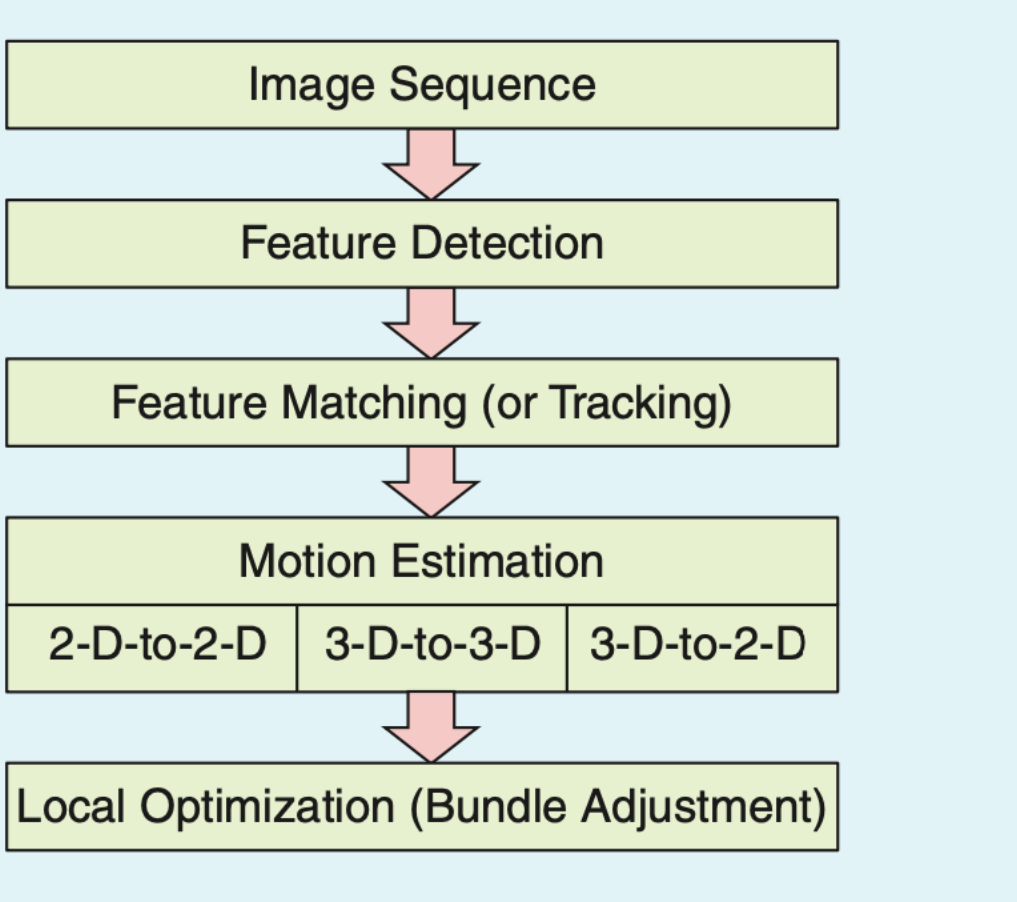


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Recap

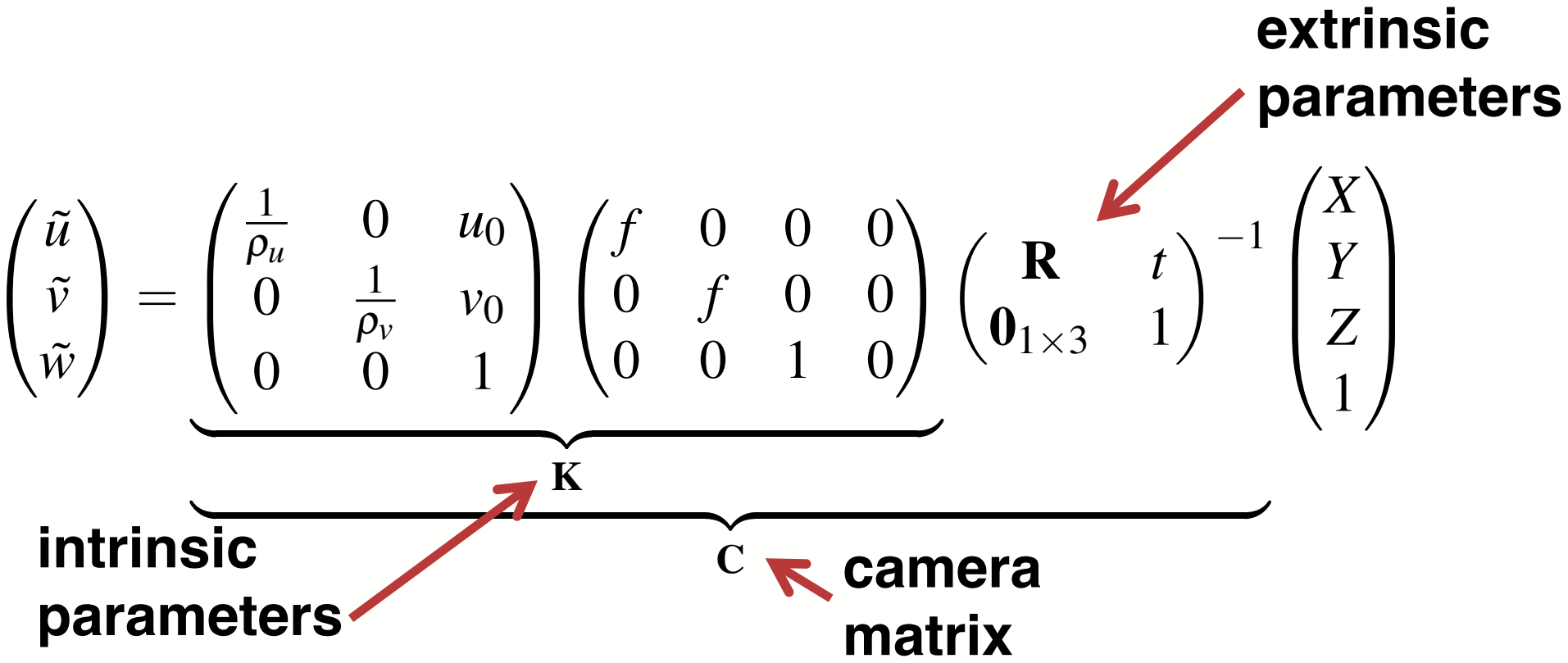
Recap – What we learned so far



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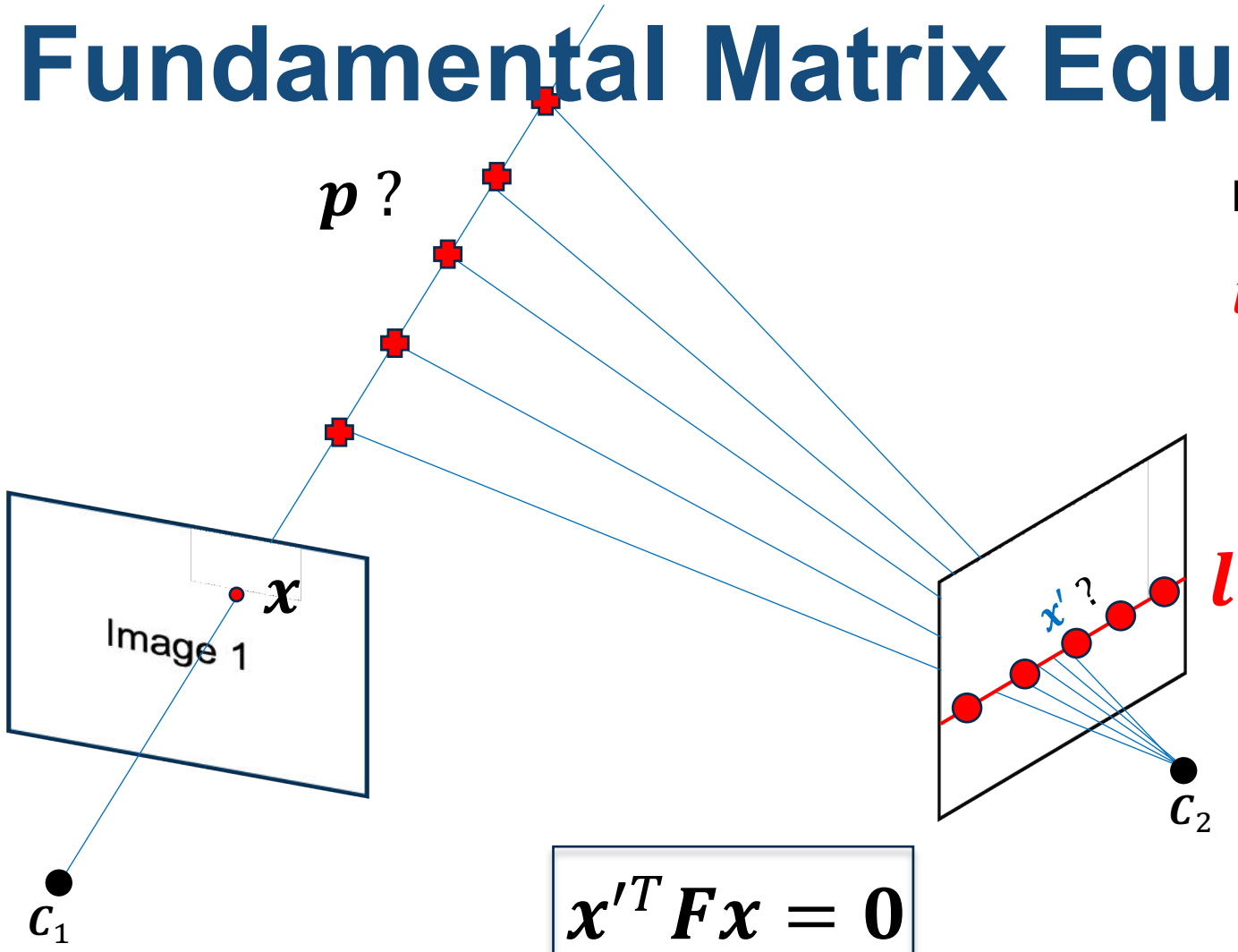
Corke, P. | Reproduced with permission from Springer Science, & Business Media

Complete camera model



Slide / Illustrations by Peter Corke

The Fundamental Matrix Equation



F is the **Fundamental Matrix** (3x3)

l is called the **Epipolar Line** of x.

$$l = Fx$$

x' must be on l!

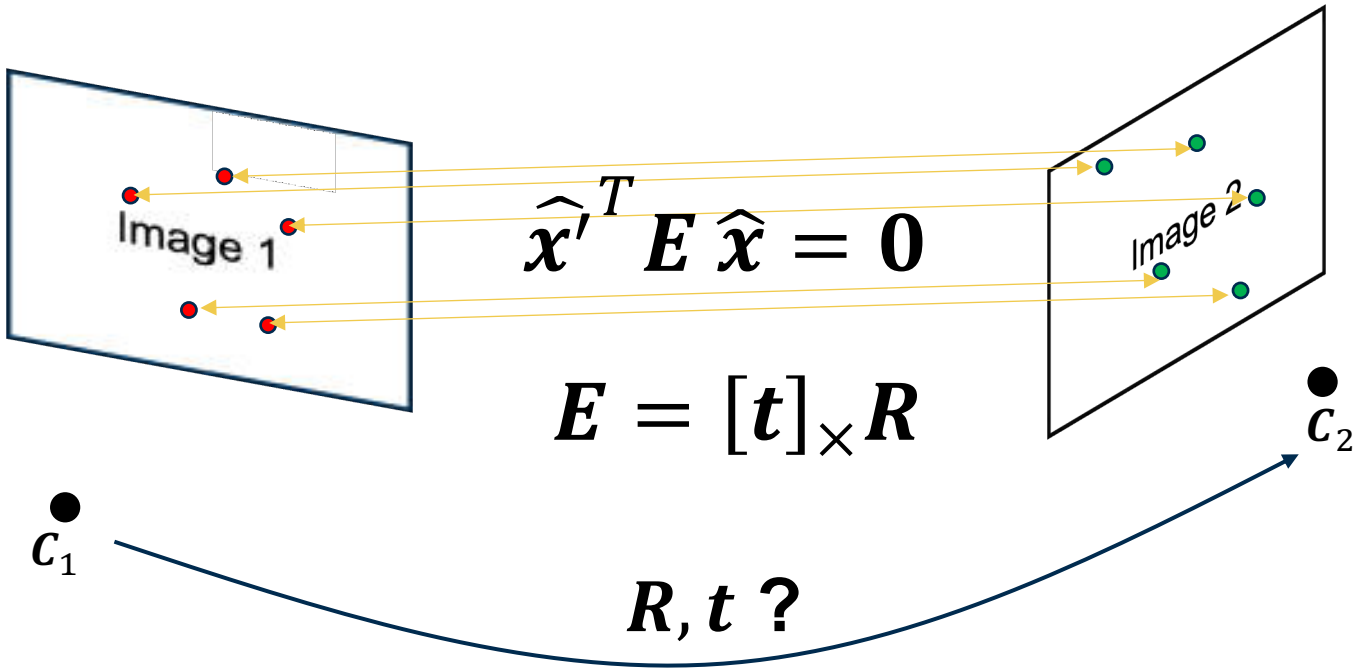
$$x'l = 0$$

Remember from the discussion of homogeneous coordinates!

$$x'^T Fx = 0$$

This is true for **any x** and corresponding **x'**!

Motion from 2D-2D Correspondences

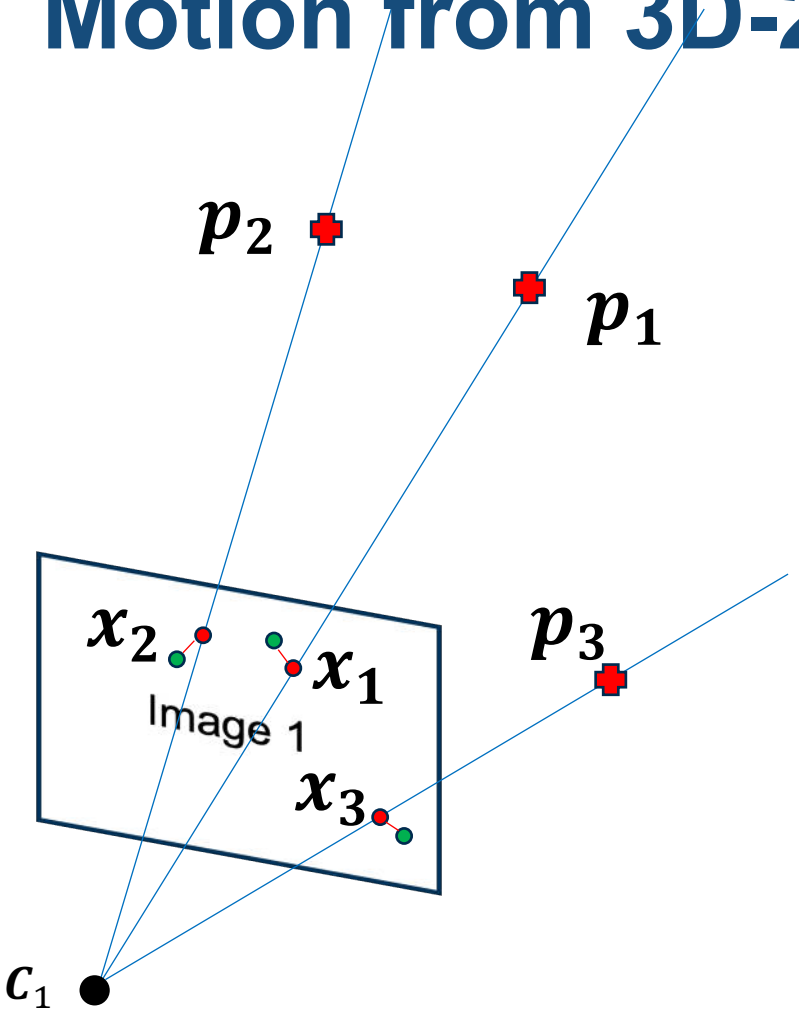


- Find interest points in both images.
- Match them.
- Calculate Essential Matrix using `findEssentialMat()`
- Find **R** and **t** using `recoverPose(E, points1, points2)`

- **R** is correct, but **t** is only the direction of the translation!
- The translation distance cannot be recovered without additional information.

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PnP: Motion from 3D-2D Correspondences



Component	PnP
Parameters	$\boldsymbol{R}, \boldsymbol{t}$
Prediction	projected pixel $\boldsymbol{K}(\boldsymbol{R} \cdot \boldsymbol{p}^{i}{}_{\tau-1} + \boldsymbol{t})$
Residual / Error	Reprojection Error $\boldsymbol{x}_{\tau}^{i} - \boldsymbol{K}(\boldsymbol{R} \cdot \boldsymbol{p}^{i}{}_{\tau-1} + \boldsymbol{t})$
Objective	sum of squared reprojection errors $\sum_{i} \| \cdots \|^{2}$

$$\mathrm{argmin}_{\boldsymbol{R},\boldsymbol{t}} \sum_{i} \left\| \boldsymbol{x}_{\tau}^{i} - \boldsymbol{K}(\boldsymbol{R} \cdot \boldsymbol{p}^{i}{}_{\tau-1} + \boldsymbol{t}) \right\|^{2}$$

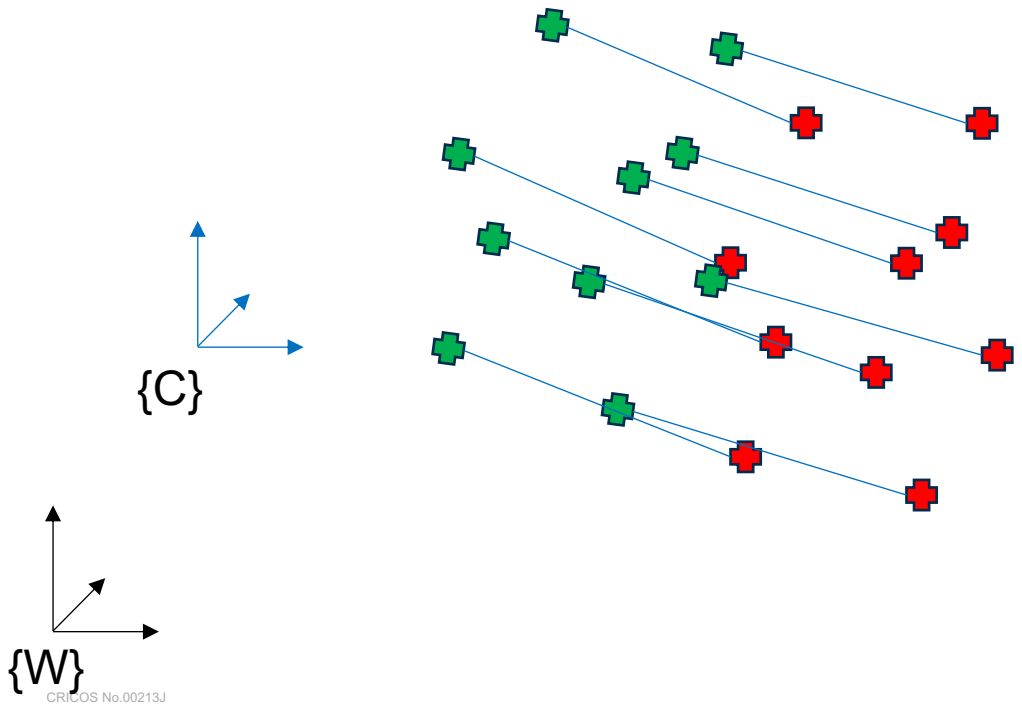
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Motion from 3D-3D Correspondences

$$\text{argmin}_{\boldsymbol{R},\boldsymbol{t}} \sum_i \left\| \boldsymbol{x}^i{}_{\tau} - (\boldsymbol{R} \cdot \boldsymbol{x}^i{}_{\tau-1} + \boldsymbol{t}) \right\|^2$$

Idea:

- Find the **R** and **t** that mininise the Euclidean distance between both sets of corresponding 3D points.

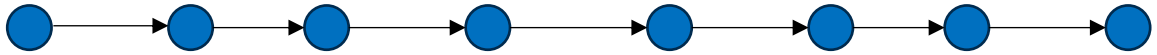


Component	3D-3D Motion Estimation
Parameters	$\boldsymbol{R}, \boldsymbol{t}$
Prediction	Transformed 3D point $\boldsymbol{R} \cdot \boldsymbol{x}^i{}_{\tau-1} + \boldsymbol{t}$
Residual / Error	Euclidean Distance in 3D $\boldsymbol{x}^i{}_{\tau} - (\boldsymbol{R} \cdot \boldsymbol{x}^i{}_{\tau-1} + \boldsymbol{t})$
Objective	sum of squared errors $\sum_i \| \cdots \|^2$

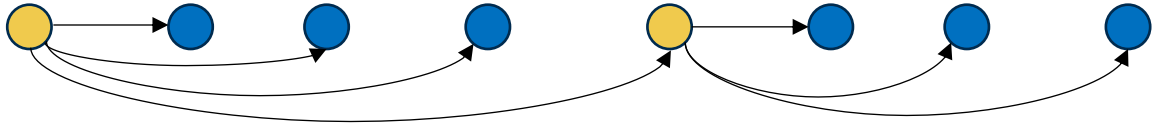
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Keyframes

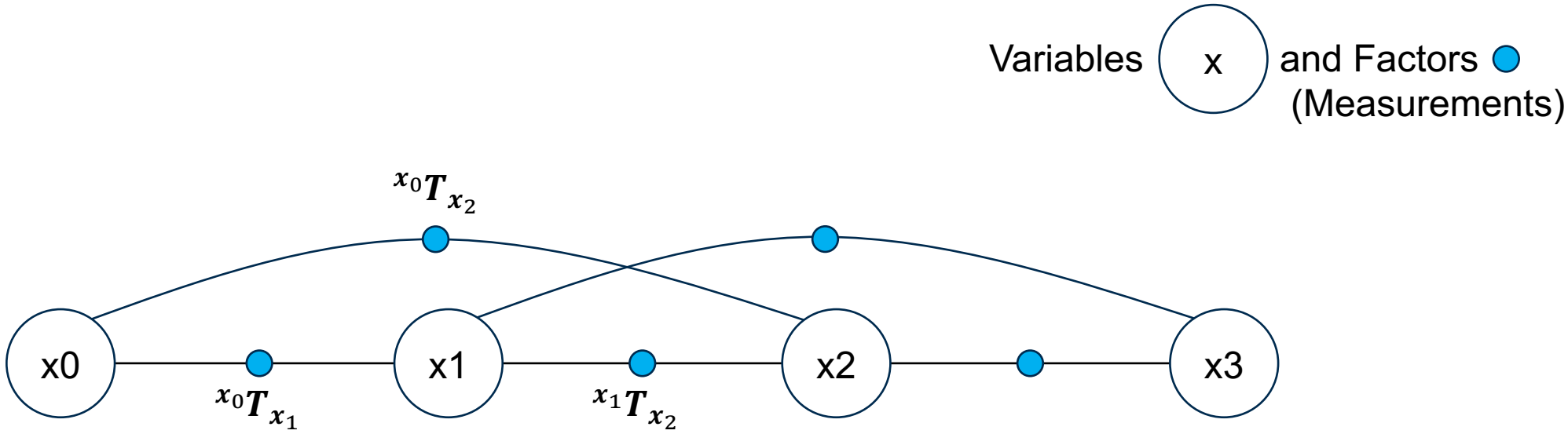
- Don’t estimate the motion between every consecutive image pair ${}^{C_{t-1}}\boldsymbol{T}_{C_t}$



- Instead, estimate the motion relative to the last keyframe: ${}^{K_i}\boldsymbol{T}_{C_t}$



Pose Graph / Factor Graph

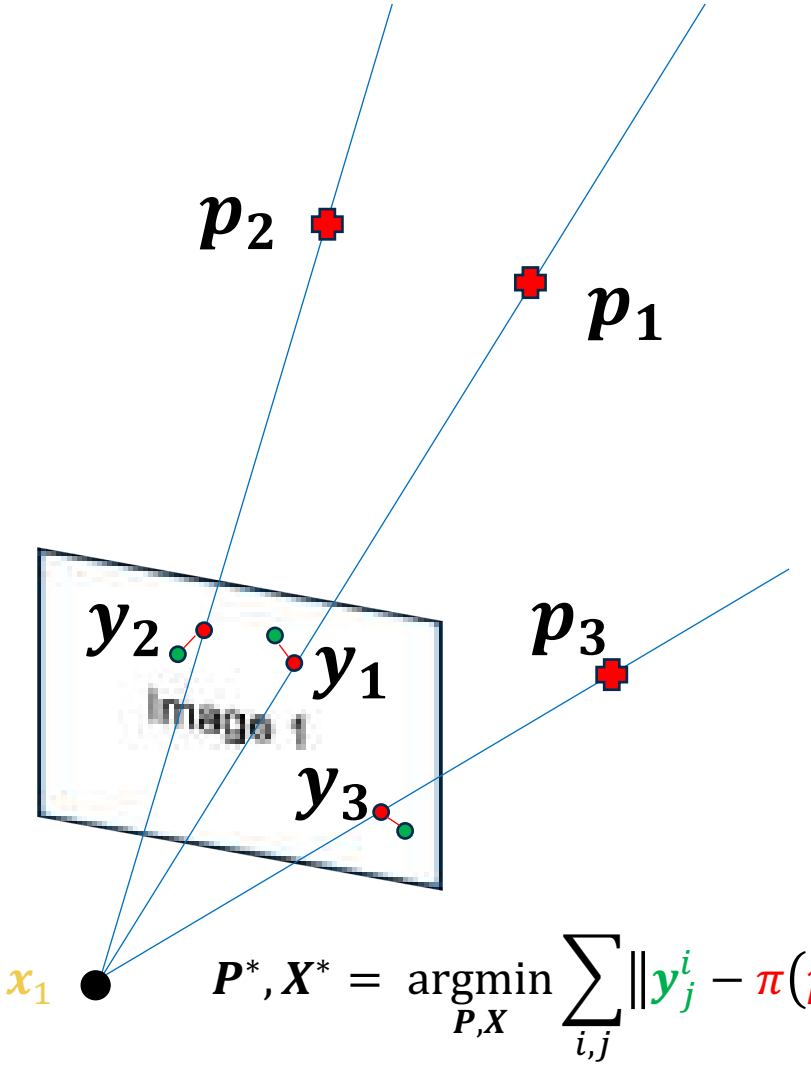


Solve by formulating it as an **optimisation problem** over the set of poses $\boldsymbol{x_i}$.

$$\boldsymbol{X}^* = \underset{\boldsymbol{X}}{\operatorname{argmin}} f(\boldsymbol{X}, \boldsymbol{T})$$

Bundle Adjustment



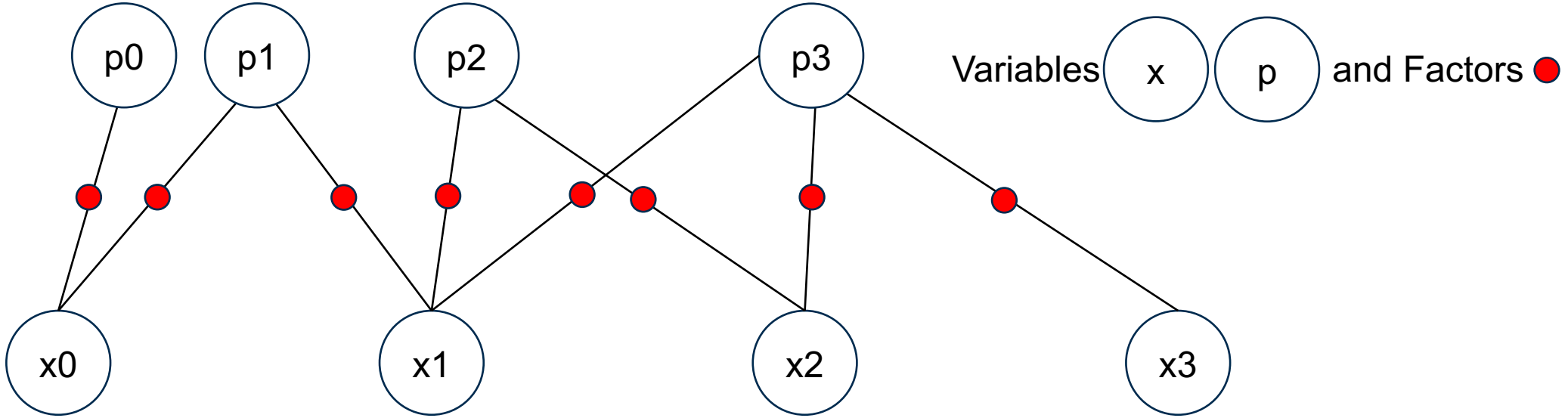


$$P^*, X^* = \underset{P,X}{\operatorname{argmin}} \sum_{i,j} \left\| y_j^i - \pi(p_j, x_i, K) \right\|^2$$

*"What are the optimal camera poses **x** and 3D point coordinates **p**, given the observations **y**?"*

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Bundle Adjustment



Optimise an error function (reprojection error) jointly over camera poses **x** and 3D point coordinates **p**.

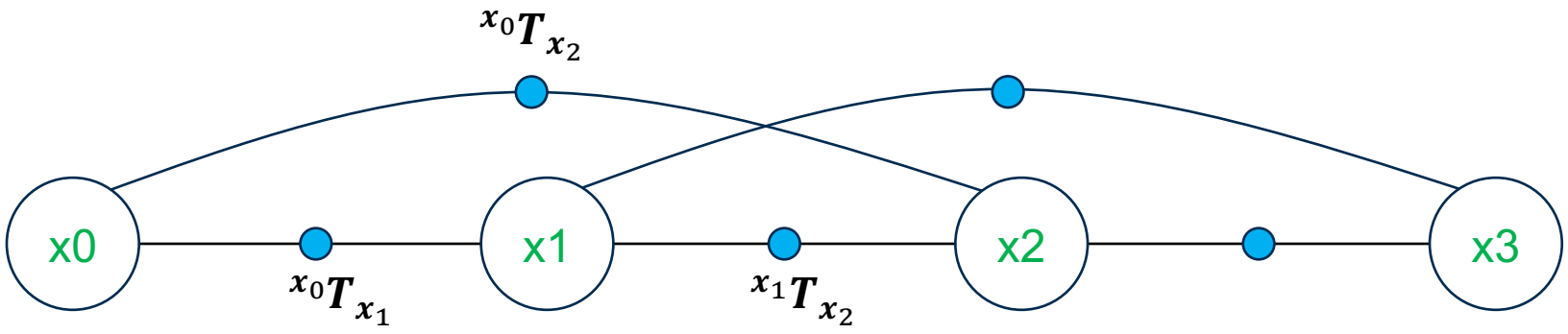
*"What are the optimal camera poses **x** and 3D point coordinates **p**, given the observations **y**?"*

$$\boldsymbol{P}^*, \boldsymbol{X}^* = \underset{\boldsymbol{P},\boldsymbol{X}}{\operatorname{argmin}} \sum_{i,j} \left\| \boldsymbol{y}_j^i - \pi\left(\boldsymbol{p}_j, \boldsymbol{x}_i, \boldsymbol{K}\right) \right\|^2$$

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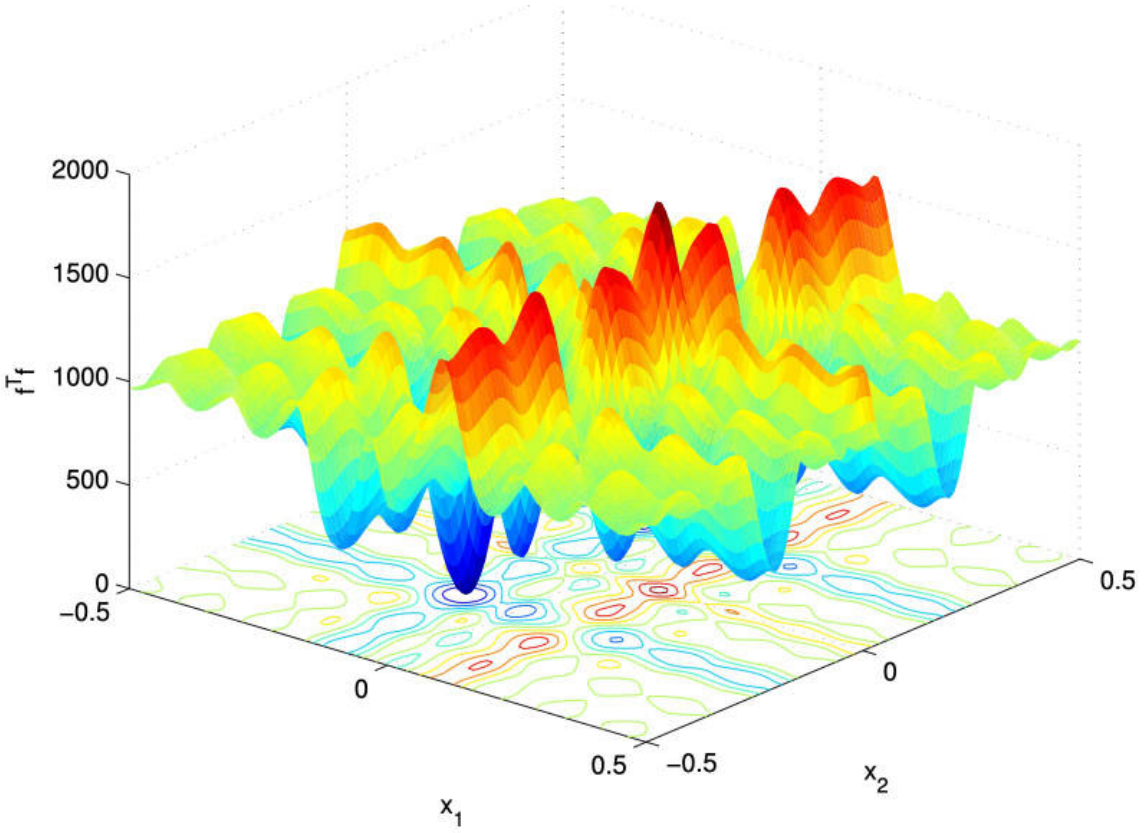
Factor Graphs

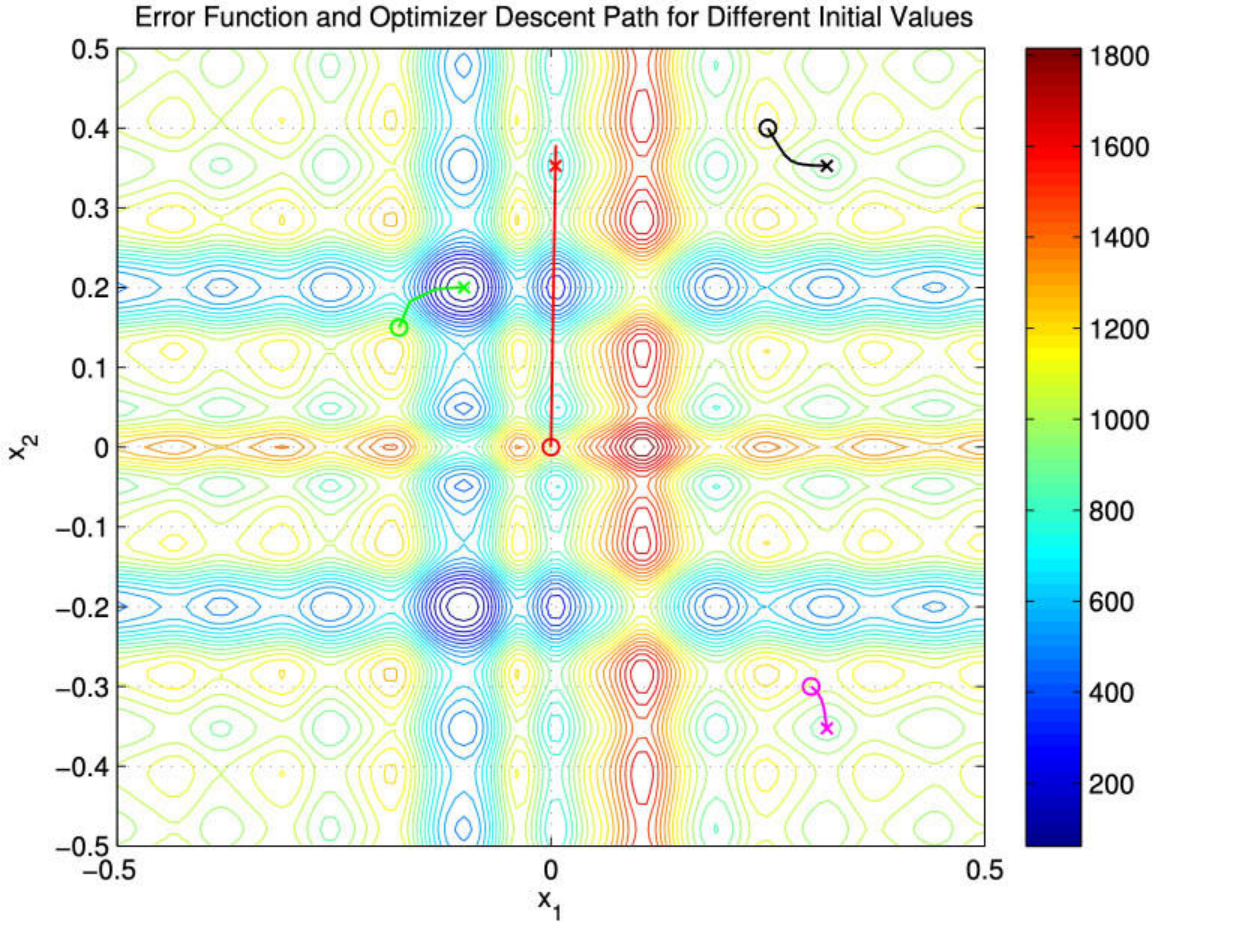
$$\mathbf{X}^* = \underset{\boldsymbol{x}}{\operatorname{argmin}} \sum_{i,j} \left\| \boldsymbol{x}_i - {}^{i}\boldsymbol{T}_j \cdot \boldsymbol{x}_j \right\|^2_{\boldsymbol{\Sigma}_{i,j}}$$



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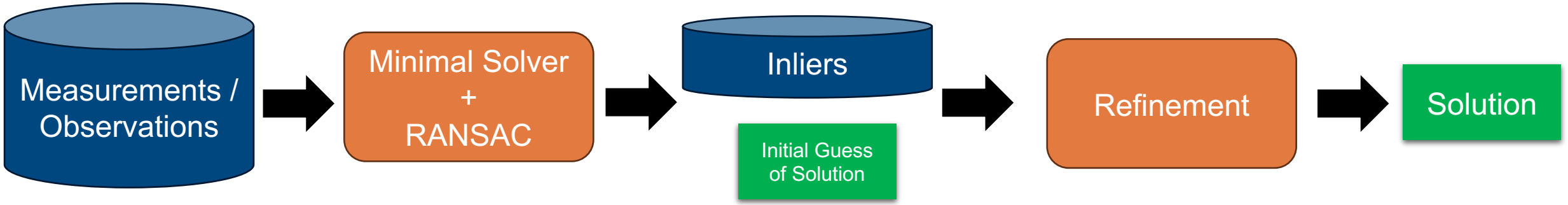
Solution depends on initial guess!





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A very typical pipeline in Computer Vision



- Typically a lot of data but contains outliners
- *e.g. matched features*

Identifies outliers

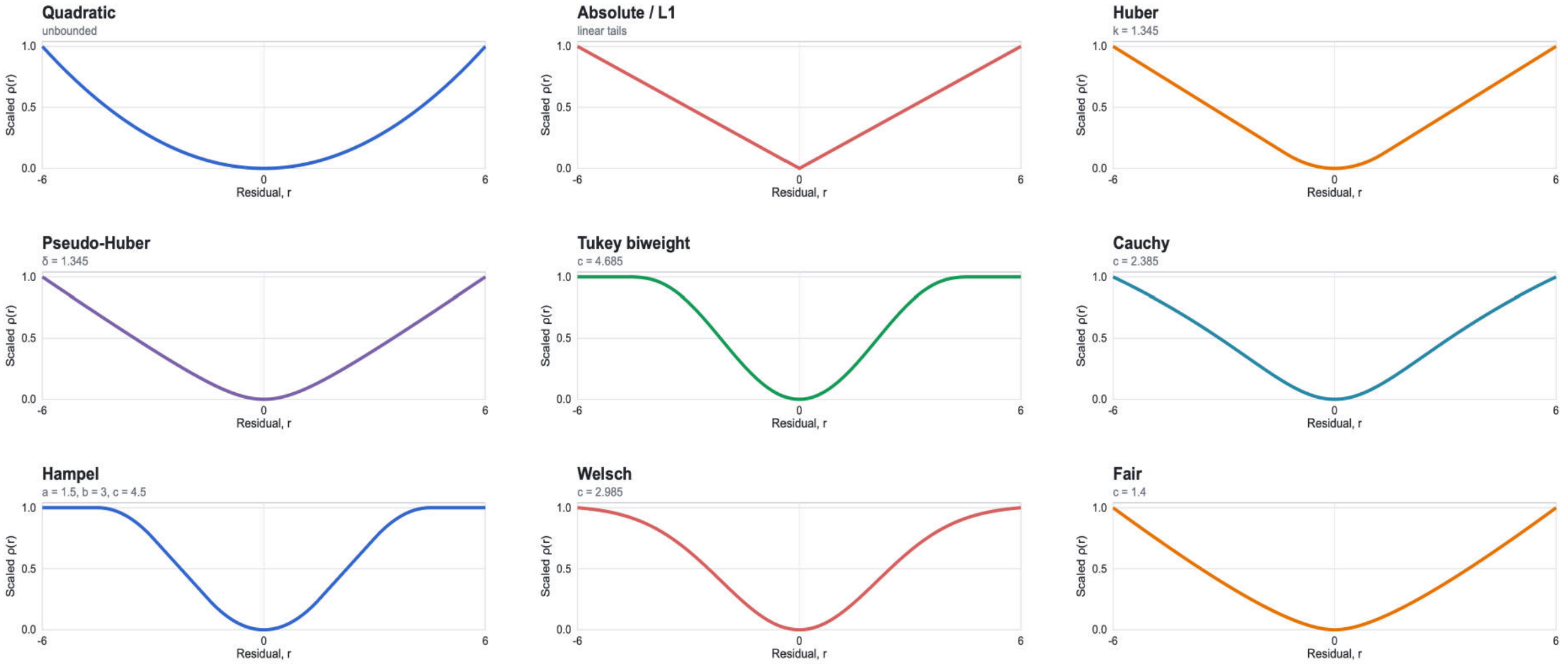
- Reduced set of inliers, guaranteed* to be free of outliers
- Solution based only on minimal set
- *e.g. 8 point pairs for F, 3 point pairs for P3P*

- Typically iterative (nonlinear) Least Squares
- Requires initial guess
- Very vulnerable to outliers

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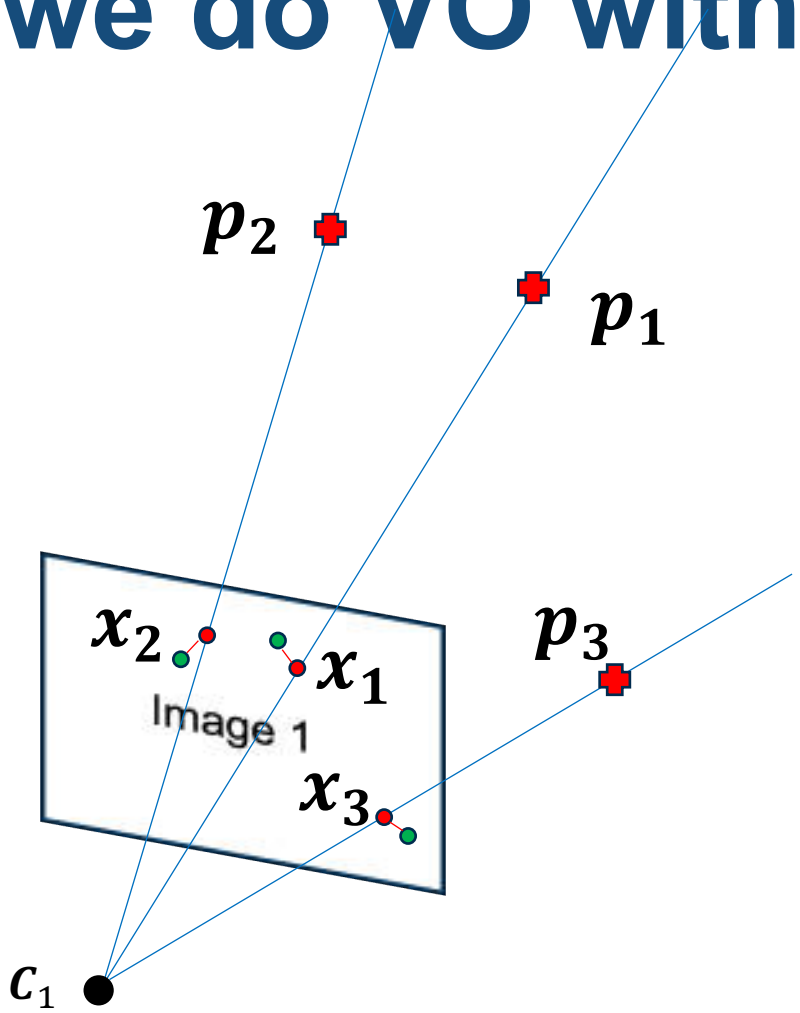
M-estimator loss functions

Residual range −6 to 6; each loss independently scaled to a maximum of 1



Direct vs Indirect Methods

Can we do VO without Features?



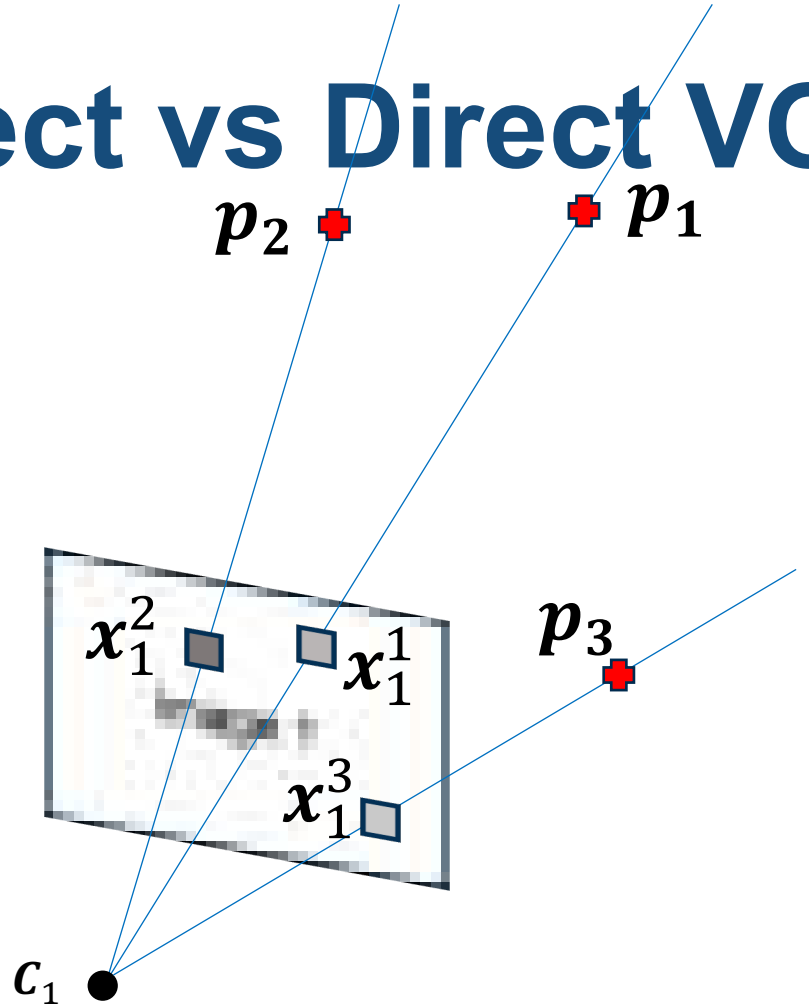
All Visual Odometry methods we talked about are build on feature correspondences.

- 3D-2D methods like PnP optimise reprojection error $\text{argmin}_{R,t} \sum_i \left\| x_\tau^i - K(R \cdot p^i{}_{\tau-1} + t) \right\|^2$
- 2D-2D is based on Essential matrix, formed from correspondences

Can we do VO without features?

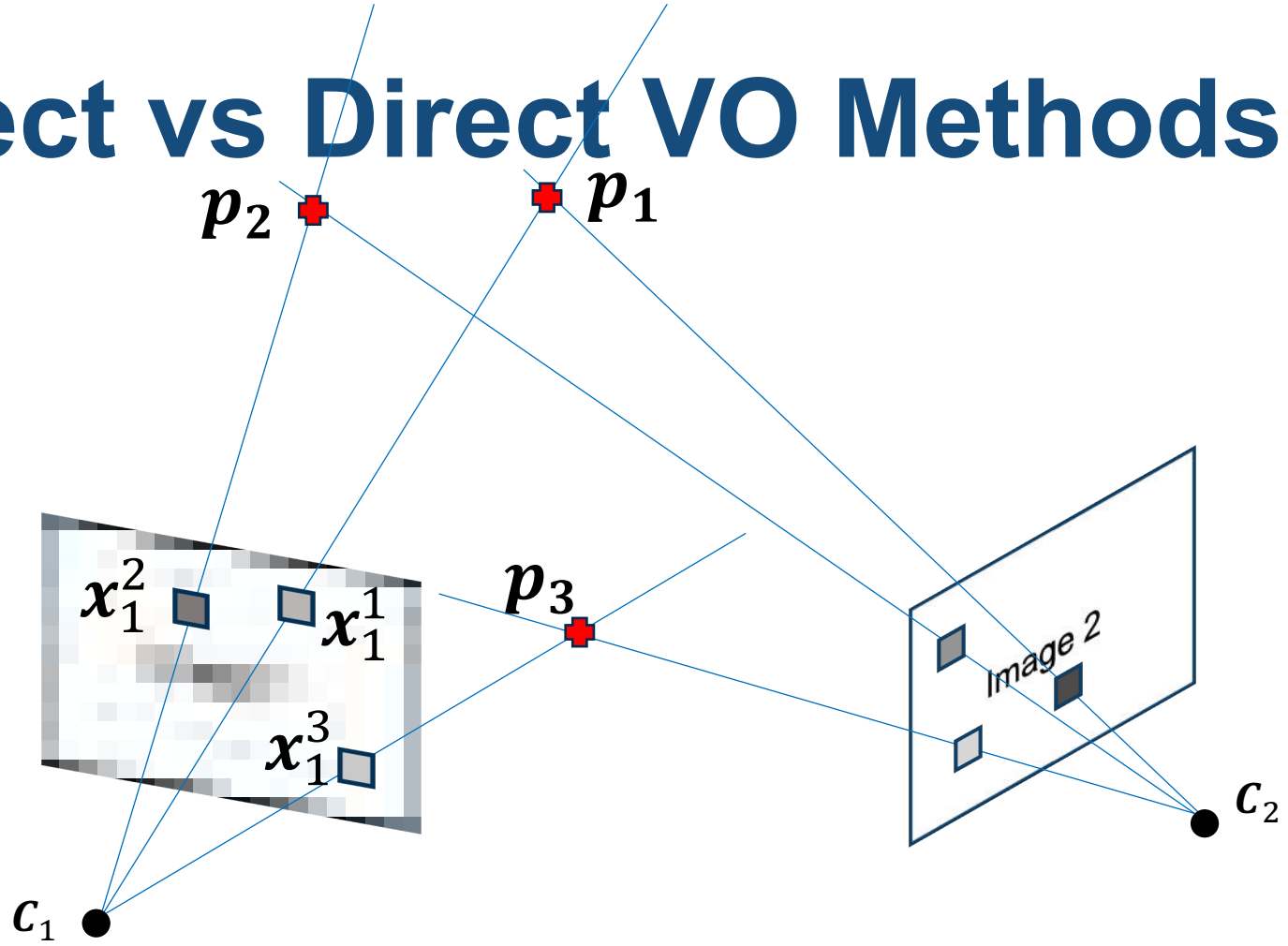
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Indirect vs Direct VO Methods



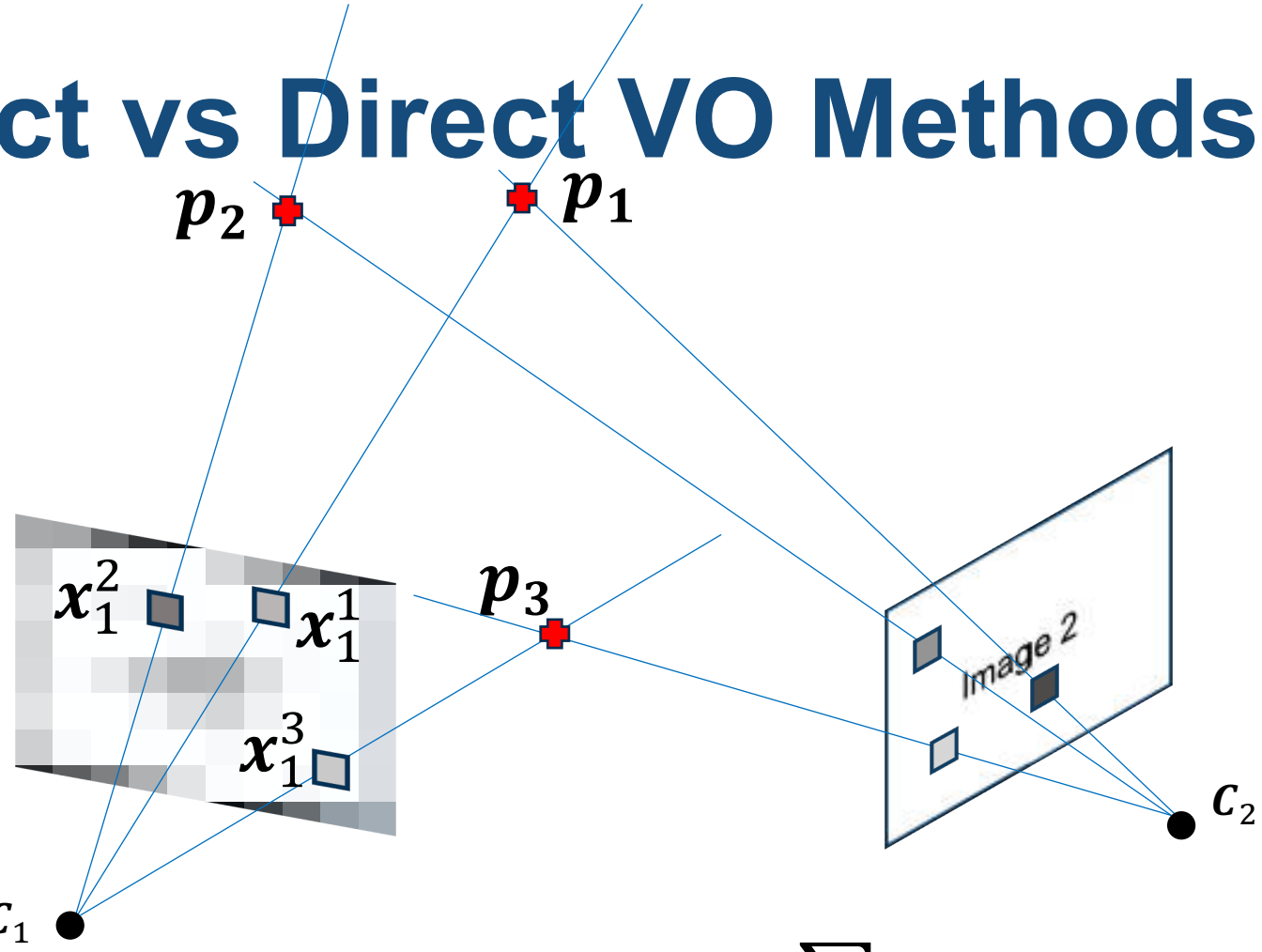
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Indirect vs Direct VO Methods



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Indirect vs Direct VO Methods

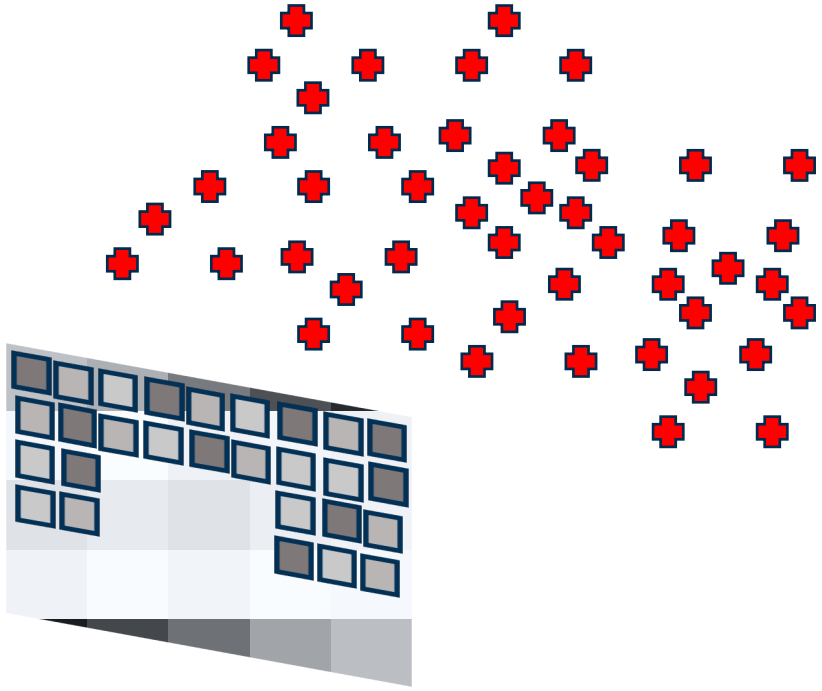


Direct Method:
minimise **photometric** error

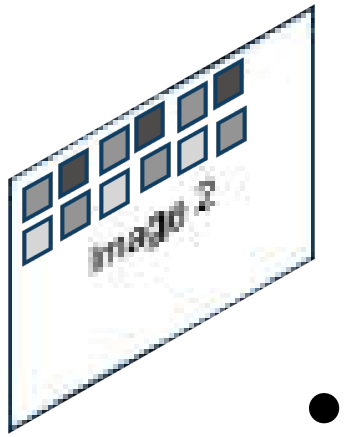
$$R^*, t^* = \text{arg}min_{R,t} \sum_i \left\| \mathcal{I}_{\tau-1}\left(x^i_{\tau-1}\right) - \mathcal{I}_\tau(K(R \cdot p^i + t)) \right\|^2$$

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Indirect vs Direct VO Methods



We can also do this as a dense method, with a Bundle Adjustment objective!



C_2

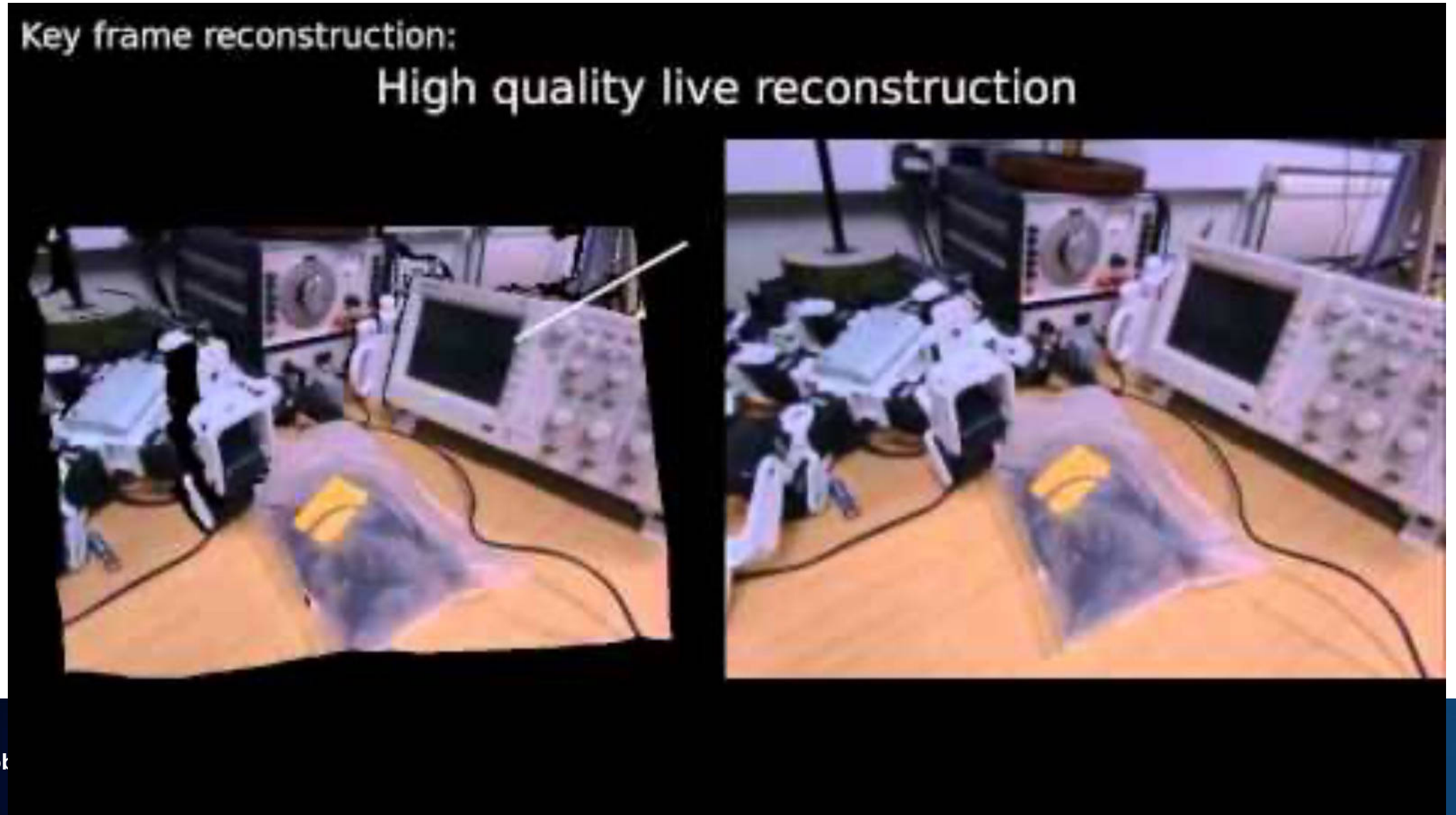
C_1

Direct Method:
minimise **photometric** error

$$\boldsymbol{R}^*, \boldsymbol{t}^*, \boldsymbol{p}^{\boldsymbol{i}*} = \mathrm{argmin}_{\boldsymbol{R},\boldsymbol{t}} \sum_i \left\| \mathcal{I}_{\tau-1}\left(\boldsymbol{x}^i_{\tau-1}\right) - \mathcal{I}_\tau(\boldsymbol{K}(\boldsymbol{R}\cdot\boldsymbol{p}^{\boldsymbol{i}} + \boldsymbol{t})) \right\|^2$$

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DTAM – Dense Tracking and Mapping



Indirect vs Direct VO Methods

Direct methods	Feature-based methods
Minimise photometric error	Minimise geometric reprojection error
Use pixel intensities	Use extracted feature locations
Correspondence is implicit through projection	Correspondence is established through descriptor matching
Can produce semi-dense or dense maps	Usually produce sparse landmark maps
Often high subpixel precision	Usually have a larger convergence basin
Sensitive to illumination and camera modelling	More tolerant of photometric changes
Require good initial guess	Better suited to large motions and relocalisation

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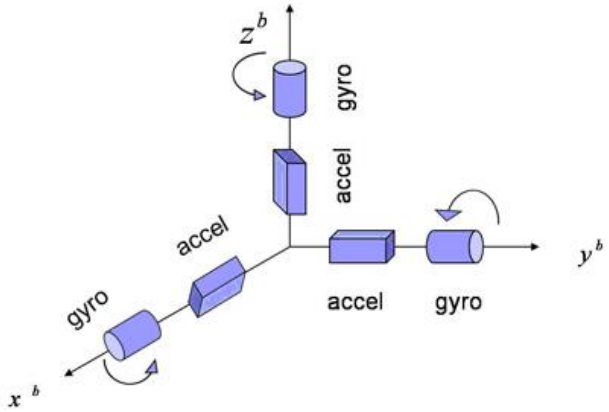


Visual-Inertial
Odometry / SLAM

Visual Inertial Odometry / SLAM

What is an IMU?

- **I**nertial **M**easurement **U**nit
  - Gyroscope: Angular velocity
  - Accelerometer: Linear Accelerations



Inertial Measurement Unit
3 accelerometers, 3 gyroscopes

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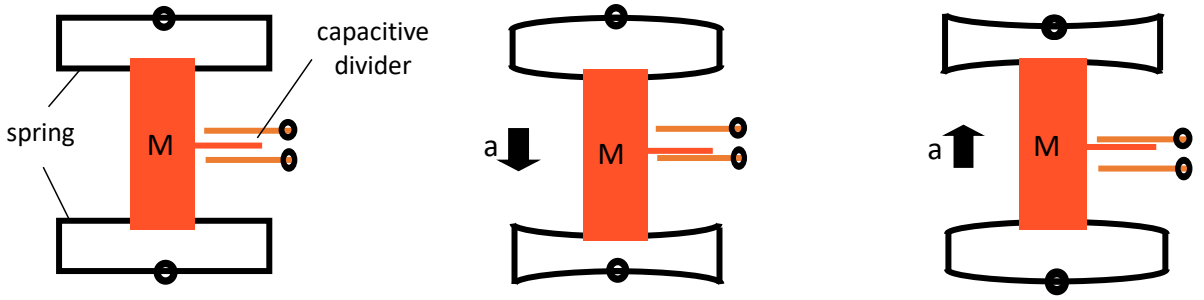
Slide courtesy of Davide Scaramuzza

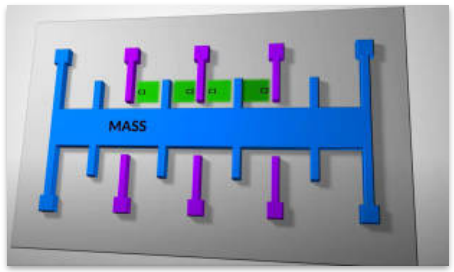
Visual Inertial Odometry / SLAM

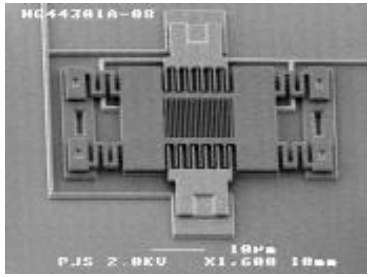
MEMS Accelerometer

(MEMS = Micro-electromechanical System)

A spring-like structure connects the device to a seismic mass vibrating in a capacitive divider. A capacitive divider converts the displacement of the seismic mass into an electric signal. Damping is created by the gas sealed in the device.









Slide courtesy of Davide Scaramuzza

Visual Inertial Odometry / SLAM

- **How does an IMU help?**
  - Monocular vision is scale ambiguous
  - Vision alone not robust enough
    - Low texture
    - Under- or over exposure, caused by low dynamic range of image sensor
    - Motion blur

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Visual Inertial Odometry / SLAM

- **Why is an IMU alone not good?**
  - Double integration (acceleration → velocity → position) of noisy signal leads to large drift over even short time
  - Often constant bias in acceleration, position error grows quadratically with time

GRADE/TIME	1 s	10 s	60 s	10 min	1 hr
Consumer	6 cm	6.5 m	400 m	200 km	39,000 km
Industrial	6 mm	0.7 m	40 m	20 km	3,900 km
Tactical	1 mm	8 cm	5 m	2 km	400 km
Navigation	<1 mm	1 mm	50 cm	100 m	10 km

Automotive,
smartphones,
and drones accelerometers

Table from Vectornav, one of the best IMU companies. Errors were computed assuming the device at rest:
https://www.vectornav.com/resources/inertial-navigation-primer/specifications--and--error-budgets/specs-inserrorbudget

Table courtesy of Davide Scaramuzza

Visual Inertial Odometry / SLAM

IMU and vision are complementary

Cameras

- **Exteroceptive sensor**: measures light energy from the environment
- × Sensitive to motion blur, HDR, texture
- ✓ Drift is bounded when observing the same scene
- ✓ Precise in slow motion
- × Limited output rate (~100 Hz)
- × Scale ambiguity in monocular setup

IMU

- **Proprioceptive sensor**: measures values internal to the system
- ✓ Insensitive to motion blur, HDR, texture
- × Drift grows unbounded regardless of the environment
- × Less precise in slow motion (low signal-to-noise ratio)
- ✓ High output rate (1,000-10,000 Hz)

- ✓ Can be used as a prior to predict next feature positions

What cameras and IMU have in common: both can be used to estimate the pose incrementally (known as dead-reckoning), which suffers from drift over time. **Solution: fuse them together to reduce drift (see later)**

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Slide courtesy of Davide Scaramuzza

IMU Measurement Model

The model measures the **angular velocity** $\tilde{\omega}_B(t)$ and **acceleration** $\tilde{a}_B(t)$ **vectors** in the body frame B:

$$\tilde{\omega}_B(t) = \omega_B(t) + b^G(t) + n^G(t)$$

$$\tilde{a}_B(t) = R_{BW}(t)(a_W(t) - g) + b^A(t) + n^A(t)$$

IMU biases + noise in body frame

Raw IMU measurements
(i.e., what you read from the sensor)

true $\boldsymbol{\omega}$ (in body frame) and true $\mathbf{a}$ (in world frame) to estimate

Notation:

- The superscript $()^G$ stands for gyroscope and $()^A$ for accelerometer
- R_{BW} is the rotation of the World frame W with respect to Body frame B
- The gravity vector g is expressed in the World frame
- Biases and noise are expressed in the body frame

Slide courtesy of Davide Scaramuzza

IMU Noise and Bias Model

- Additive, zero-mean Gaussian white noise: $n^G(t), n^A(t)$
- Biases: $b^G(t), b^A(t)$
  - The gyroscope and accelerometer biases are considered **slowly varying "constants"**. Their temporal fluctuation is modeled by saying that the derivative of the bias is a zero-mean Gaussian noise with standard deviation σ_b

$$\dot{\mathbf{b}}(t) = \sigma_b \mathbf{w}(t) \qquad \mathbf{w}(\mathrm{t}) \sim \mathbf{N}(0,1)$$

- Some facts about IMU biases:
- They change with **temperature** and **mechanical and atmospheric pressure**
- Thus, they **may also be different every time the IMU is turned on**
- Good news: **they can be estimated!** (see later)

Trawny, Roumeliotis, Indirect Kalman filter for 3D attitude estimation. Technical Report, University of Minnesota, 2005. PDF.
More info on the noise model: https://github.com/ethz-asl/kalibr/wiki/IMU-Noise-Model

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Non-linear Optimization Methods

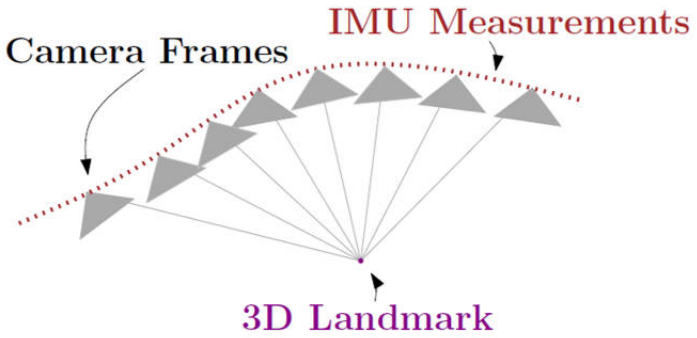
VIO is solved as a non-linear Least Square optimization problem over:

$$\{X, L, b^A, b^G\} = argmin_{\{X, L, b^A, b^G\}} \left\{ \sum_{k=1}^{N} \|f(x_{k-1}, u) - x_k\|^2_{\Lambda_k} + \sum_{k=1}^{N} \sum_{i=1}^{M} \|\pi(x_k, L^i) - z_k^i\|^2_{\Sigma_k^i} \right\}$$

IMU residuals

Reprojection residuals
(Bundle Adjustment term)

NB: it also optimizes the biases



[1] Jung, Taylor, *Camera Trajectory Estimation using Inertial Sensor Measurements and Structure from Motion Results*, International Conference on Computer Vision and Pattern Recognition (CVPR), 2001. PDF.
[2] Sterlow, Singh, *Motion estimation from image and inertial measurements*, International Journal of Robotics Research (IJRR), 2004. PDF.

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Non-linear Optimization Methods

VIO is solved as a non-linear Least Square optimization problem over:

$$\{X, L, b^A, b^G\} = argmin_{\{X, L, b^A, b^G\}} \left\{ \sum_{k=1}^{N} \|f(x_{k-1}, u) - x_k\|^2_{\Lambda_k} + \sum_{k=1}^{N} \sum_{i=1}^{M} \left\|\pi(x_k, L^i) - z_k^i\right\|^2_{\Sigma_k^i} \right\}$$

IMU residuals — *Reprojection residuals* (Bundle Adjustment term)

where

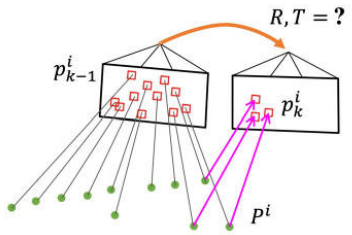
- $X = \{x_1, \dots x_N\}$: set of **state estimates** x_k (**position, velocity, orientation**) at frame times k
- $L = \{L_1, \dots, LM\}$: **3D landmarks**
- $f(x_{k-1}, u)$: **state prediction** obtained by integrating IMU measurements $u = \{\tilde{a}_j, \tilde{\omega}_j\}$
- $\pi(x_k, l_i)$: **expected measurement at state estimates** x_k from projection of landmark L_i onto camera frame I_k
- z_{i_k}: **observed features**
- Λ_k: **state covariance** from the IMU integration $f(x_{k-1}, u)$
- Σ_{i_k}: **covariance** of the 2D feature position

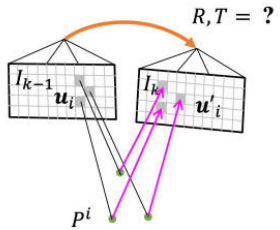
[1] Jung, Taylor, *Camera Trajectory Estimation using Inertial Sensor Measurements and Structure from Motion Results*, International Conference on Computer Vision and Pattern Recognition (CVPR), 2001. PDF.
[2] Sterlow, Singh, *Motion estimation from image and inertial measurements*, International Journal of Robotics Research (IJRR), 2004. PDF.

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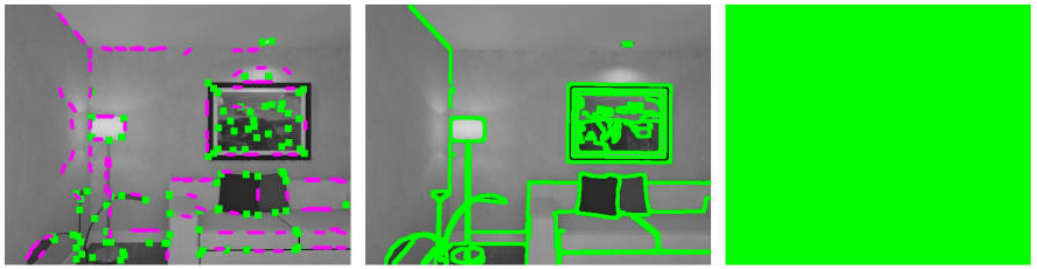
Slide courtesy of Davide Scaramuzza

Direct, Indirect, Sparse, Dense



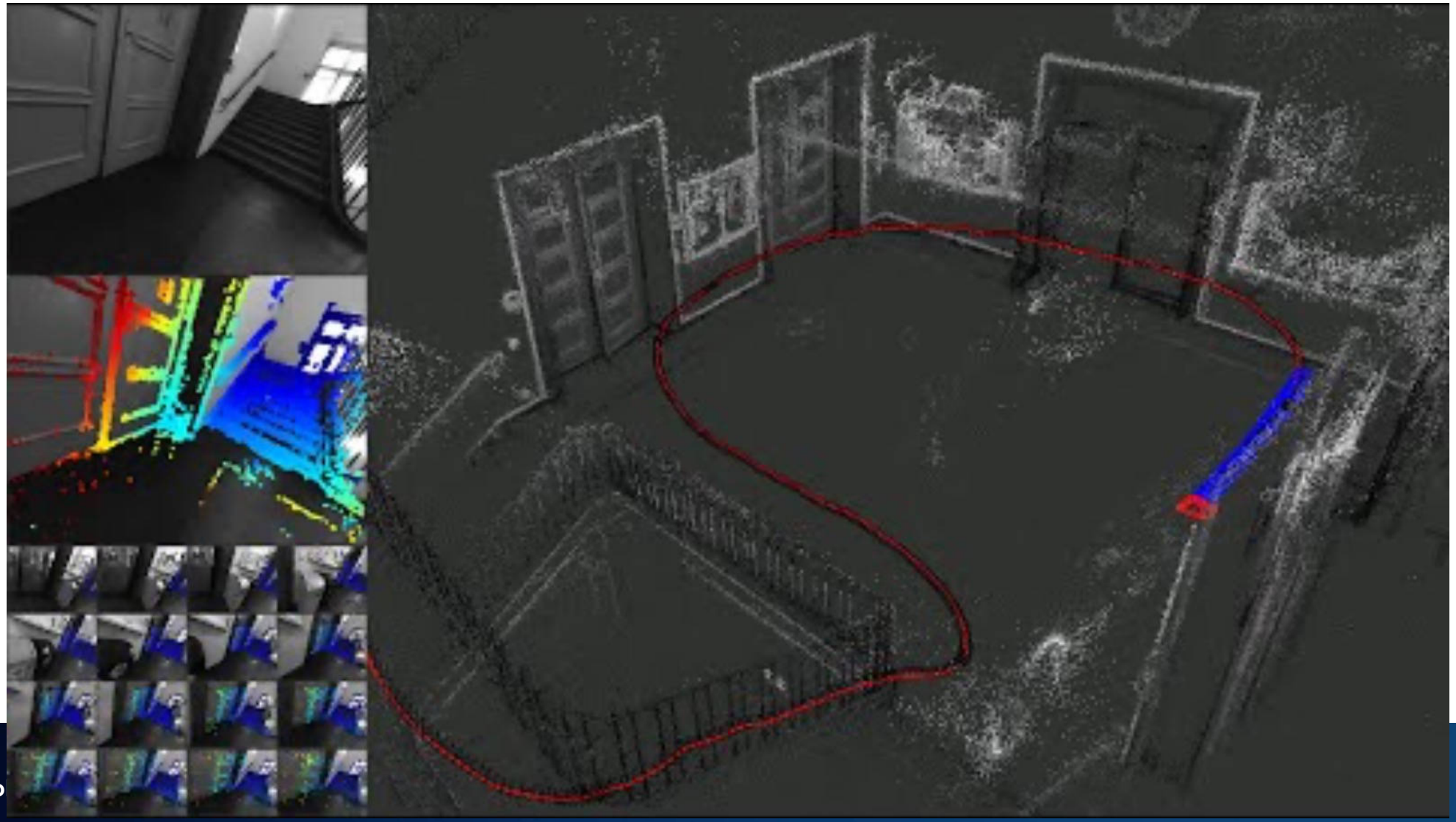


	Sparse	Dense
Indirect	• Covered in previous weeks • Reprojection Error • PTAM, ORB-SLAM, libviso, …	• 3D geometry from Dense Optical Flow (reprojection error)
Direct	• Photometric Error • DSO	• Photometric Error • DTAM

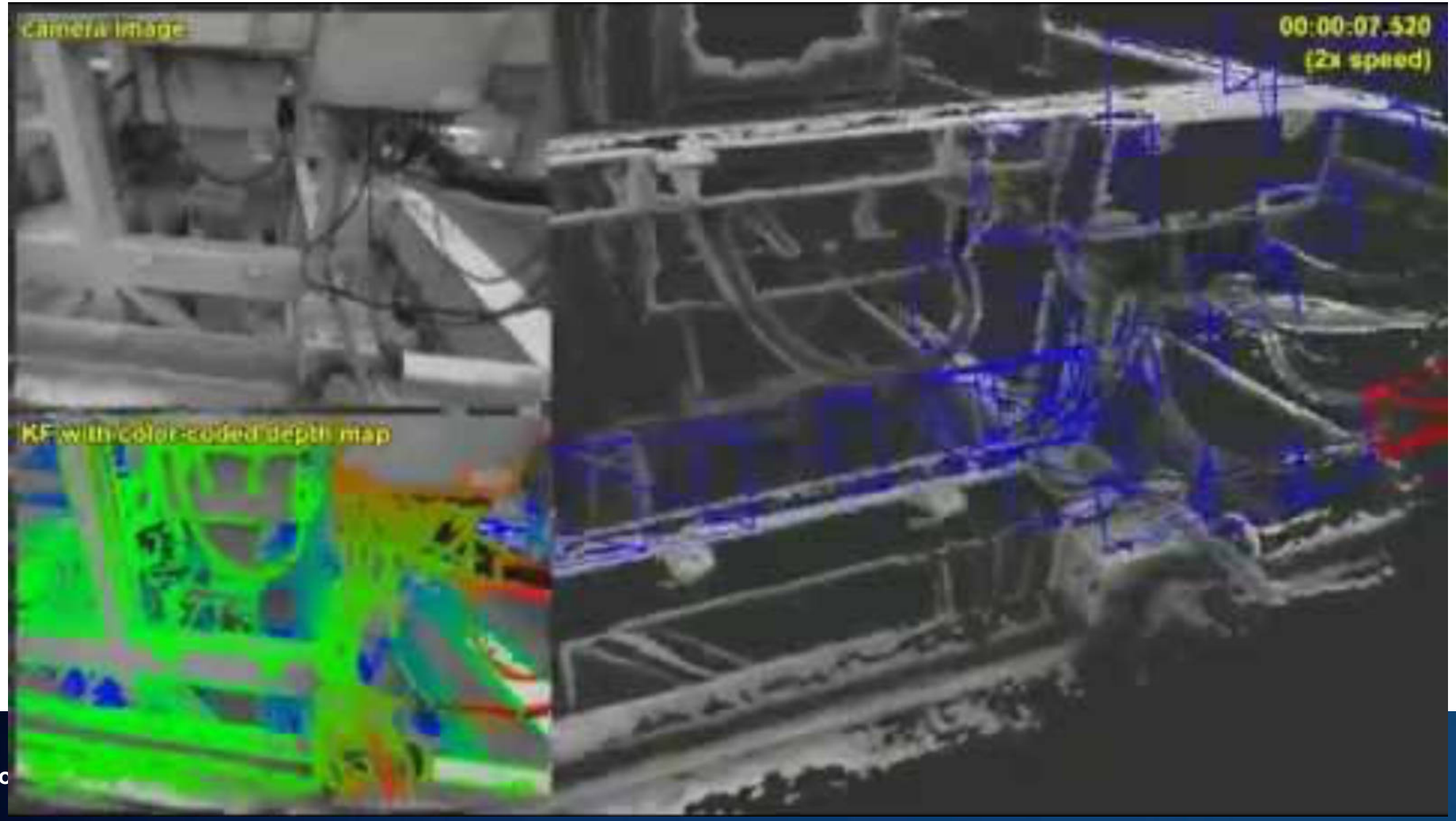


(a) Sparse (b) Semi-Dense (c) Dense

DSO – Direct Sparse Odometry



LSD-SLAM: Semi-Dense & Direct

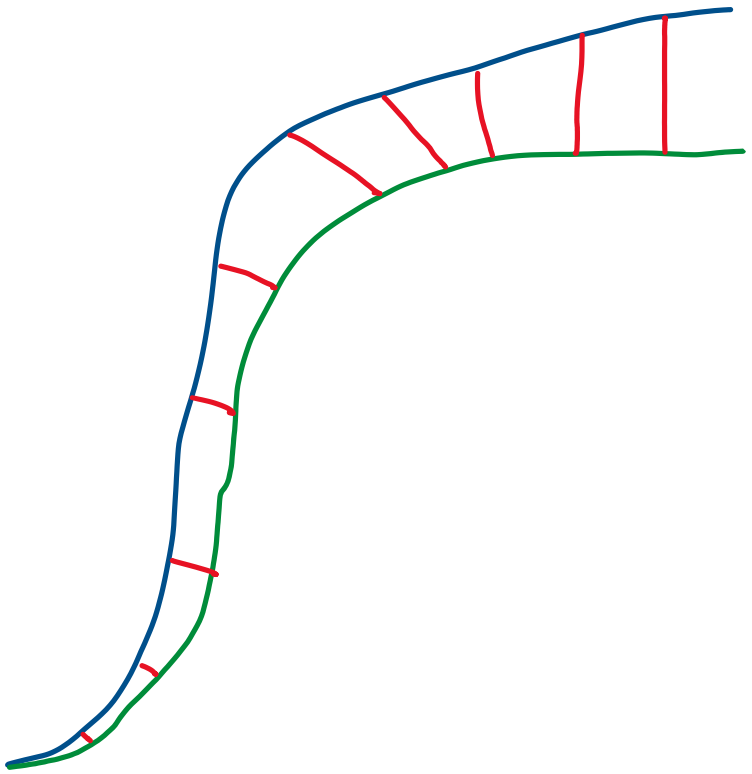


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Evaluation of Visual Odometry

Absolute Trajectory Error



- calculate the average Euclidean distance between each estimated position and the corresponding ground truth

- Clearly captures accumulated drift over the complete sequence.
- Produces an intuitive result, normally expressed in metres.
- Simple to calculate, visualise and compare.

- **Limitations**
  - Can require spatial and temporal alignment with ground truth.
  - A single RMSE value hides where and when errors occurred.
  - Sensitive to outliers (due to the averaging over the whole trajectory)
  - Does not distinguish translation drift from rotation-induced position error.
  - Can be misleading for monocular VO if scale is fitted using the complete trajectory.

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Absolute Trajectory Error with Alignment

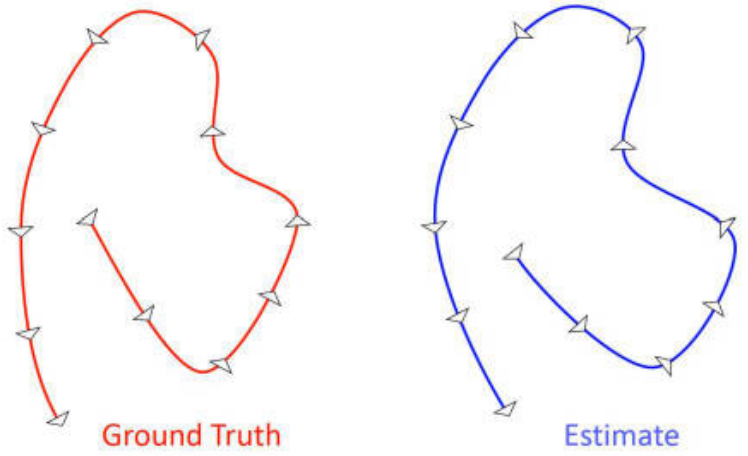
- **Step 1**: align the estimated trajectory to the ground truth from the start to the end using a similarity transformation (i.e., R, T, s) by minimizing the sum of square position errors

groundtruth positions

$$R, T, s = \underset{R,T,s}{\operatorname{argmin}} \sum_{k=0}^{n} \left\| \hat{C}_k - sRC_k - T \right\|^2$$

estimated positions

Parameters of the similarity transformation that we want to find



- **Step 2**: compute Root Mean Square Error (RMSE) after alignment:

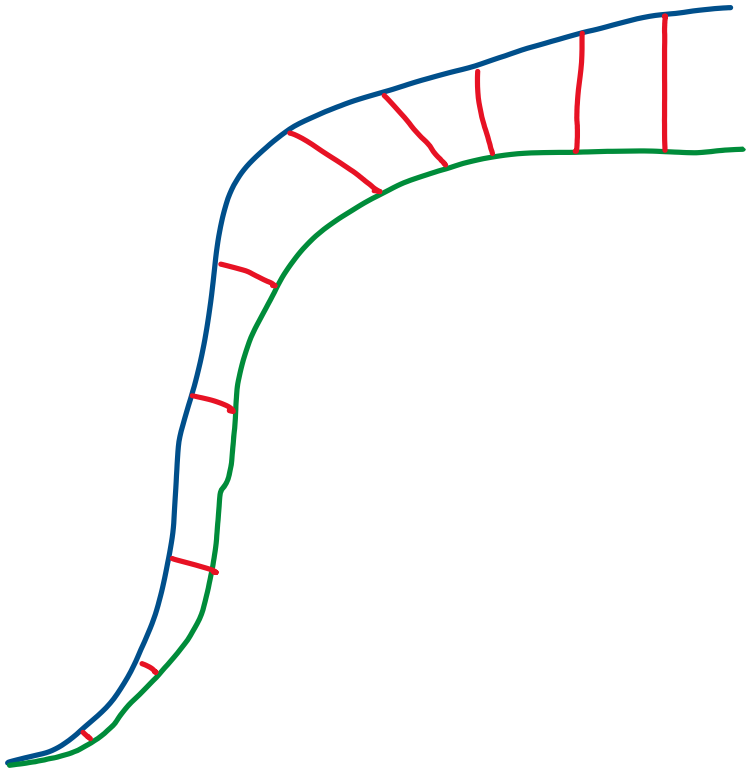
$$RMSE = \sqrt{\frac{\sum_{k=1}^{n} \left\| \hat{C}_k - sRC_k - T \right\|^2}{n}}$$

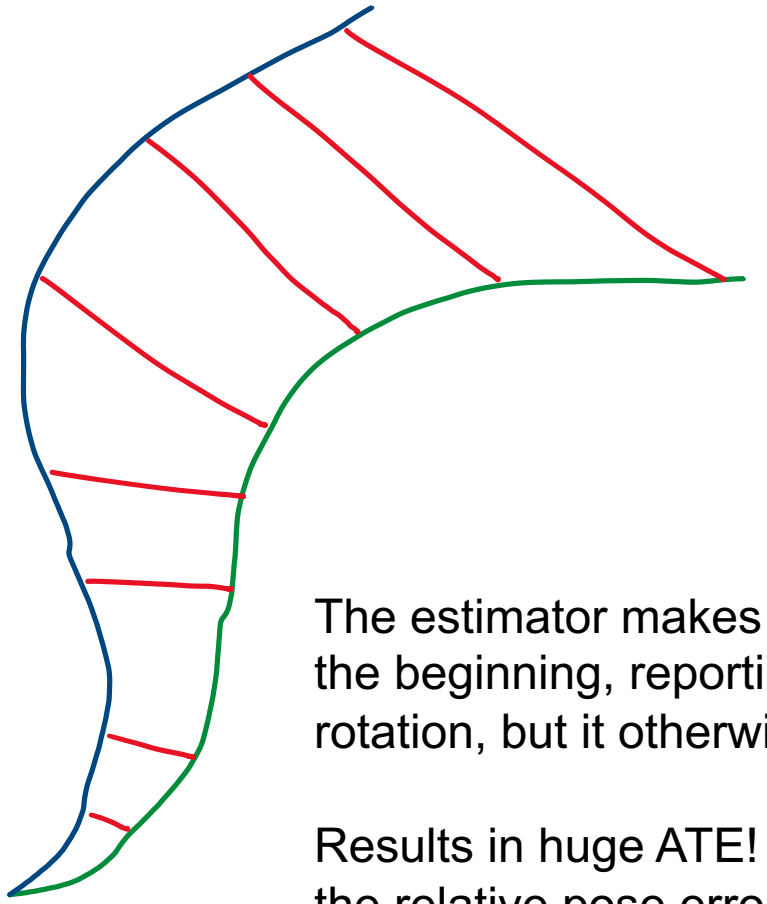
- **Pros and cons**:
  - ✓ Single-number metric
  - ✓ Captures the global error (accuracy of the global trajectory)
  - ✗ Does not capture the relative error (accuracy of the local trajectory estimate)

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Slide courtesy of Davide Scaramuzza

Absolute Trajectory Error





The estimator makes a single mistake in the beginning, reporting a wrong rotation, but it otherwise perfect.

Results in huge ATE! → Better metric is the relative pose error.

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Relative Pose Error

$$\boldsymbol{E}_i^{\mathrm{rel}} = \left[{}^{w}\boldsymbol{T}_i^{-1} \cdot {}^{w}\boldsymbol{T}_{i+1}\right]^{-1} \cdot \left[{}^{w}\widehat{\boldsymbol{T}}_i^{-1} \cdot {}^{w}\widehat{\boldsymbol{T}}_{i+1}\right] \quad \text{or} \quad \boldsymbol{E}_i^{\mathrm{rel}} = {}^{i-1}\boldsymbol{T}_i^{-1} \cdot {}^{i-1}\widehat{\boldsymbol{T}}_{i+1}$$

Relative motion between frames i and i-1

True relative motion between frames i and i-1

Difference between the two relative motions (as an SE3)

$$\mathrm{RPE}_{\mathrm{trans}} = \sqrt{\frac{1}{\mathrm{N}-1}\sum_{\mathrm{i}=1}^{\mathrm{N}-1}\left\|\mathrm{trans}\left(\boldsymbol{E}_i^{\mathrm{rel}}\right)\right\|_2^2}$$

Extract translation part from the SE3

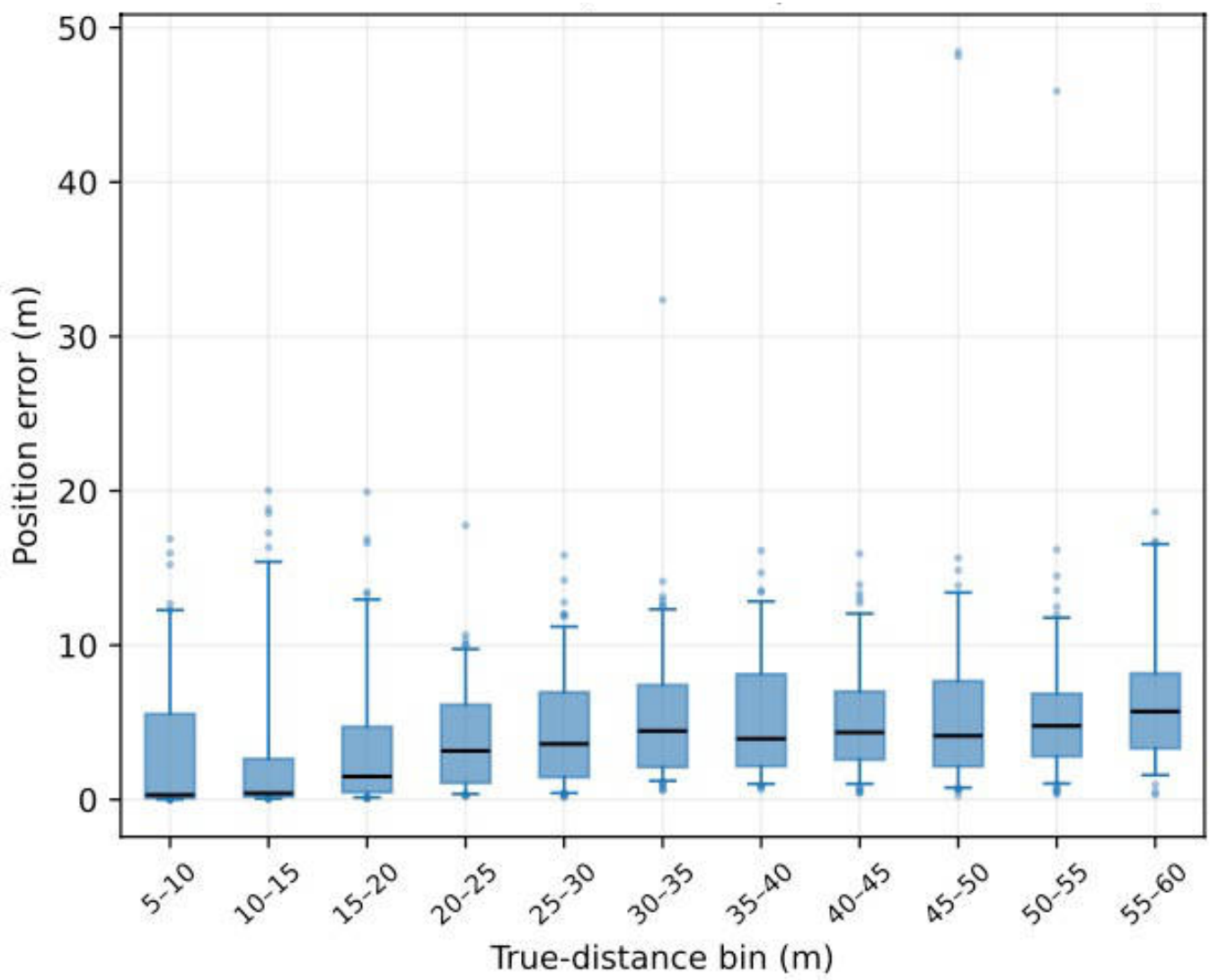
RPE is the difference in frame-to-frame relative motion, averaged over trajectory.

We can also do that for rotation component.

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Mean or Median?

- Reporting average errors can be skewed by outliers
- Consider reporting median error in addition to the average
- Boxplots reveal more of the distribution:
  - The box shows the Median (line in the middle) and 25th and 75th percentile
  - Whiskers are often 5th and 95th percentile
  - Dots or crosses are outliers outside of that range

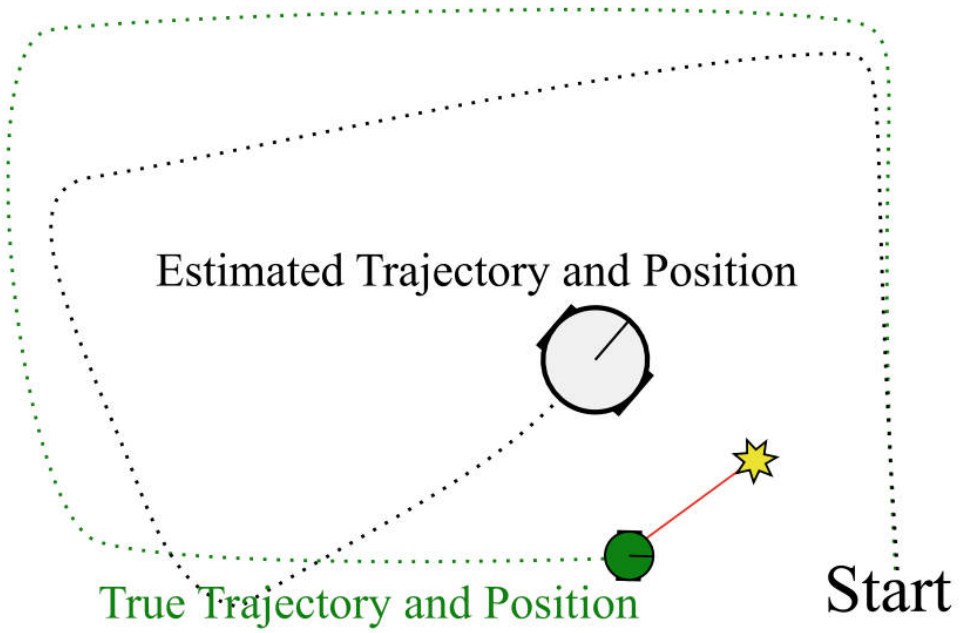


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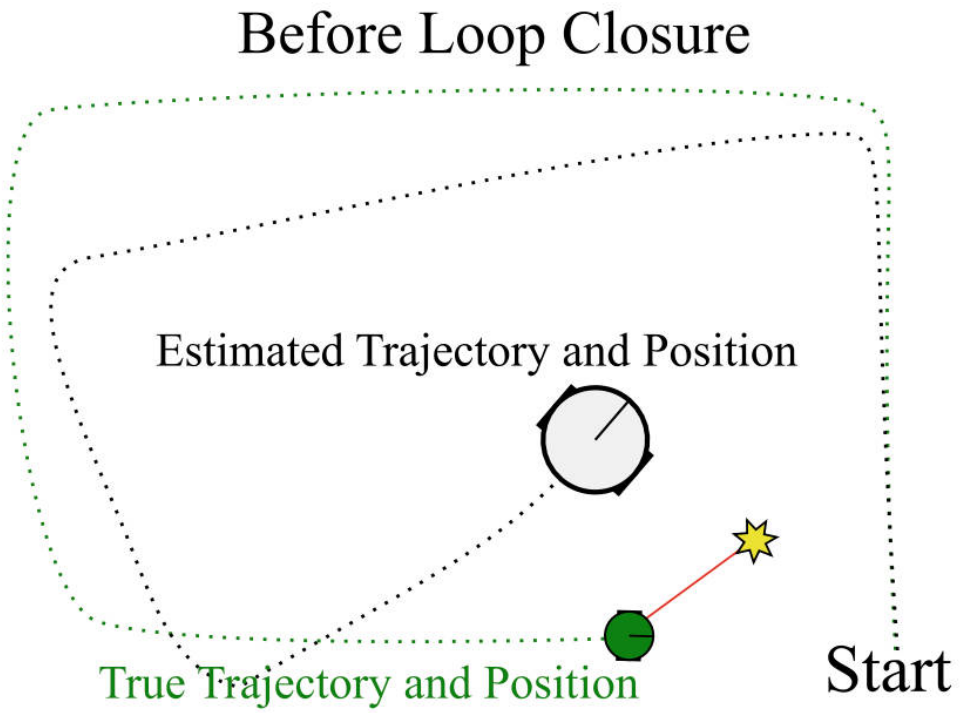
Visual Odometry vs. SLAM

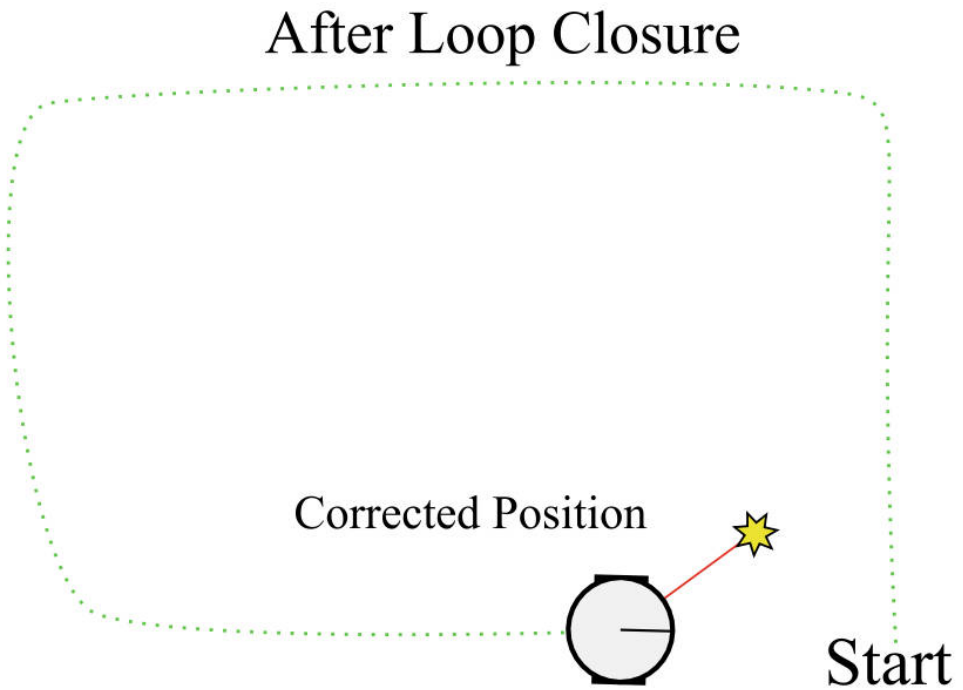
VO vs SLAM



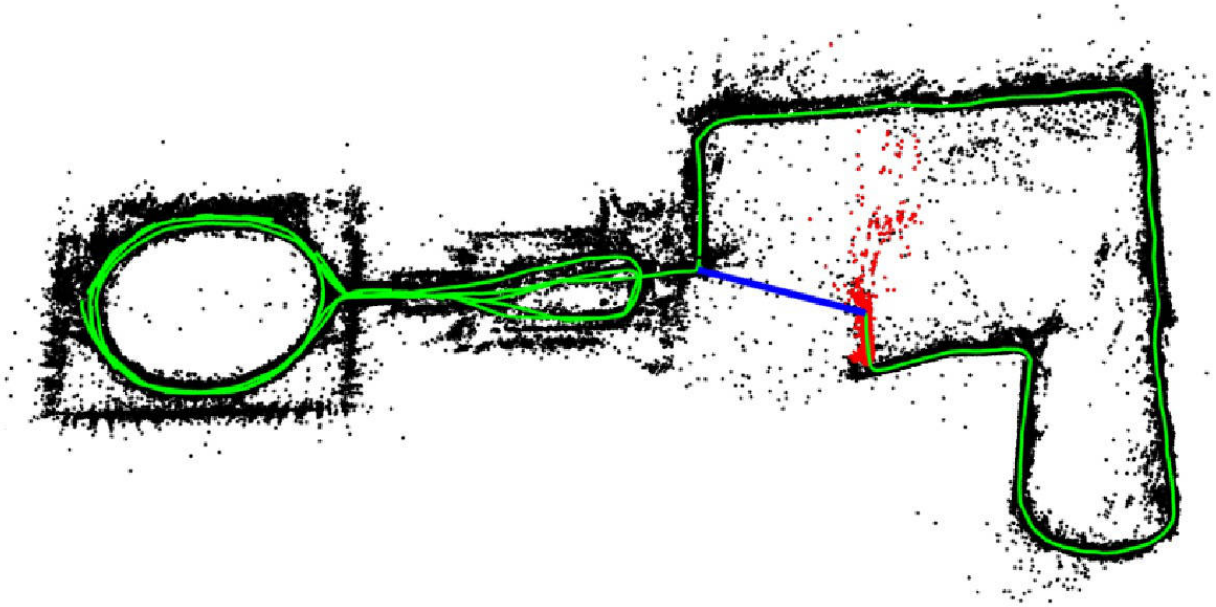
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VO vs SLAM


Before Loop Closure
Estimated Trajectory and Position
True Trajectory and Position
Start


After Loop Closure
Corrected Position
Start

VO vs SLAM



Before Loop Closure

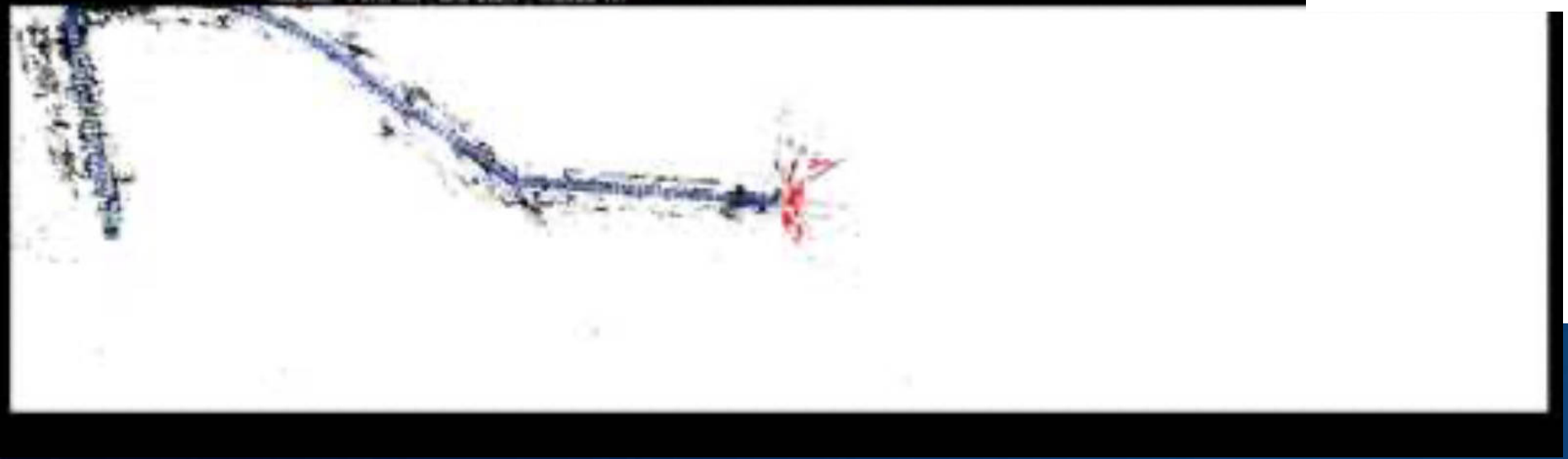


After Loop Closure

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Case Study: ORB-SLAM





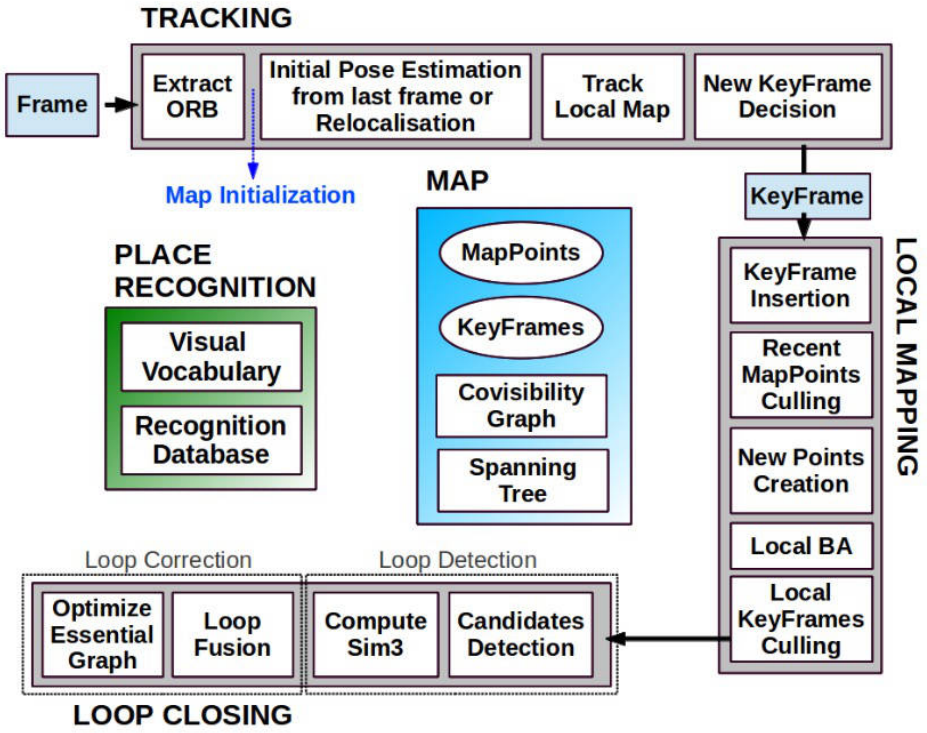


Fig. 1. ORB-SLAM system overview, showing all the steps performed by the tracking, local mapping and loop closing threads. The main components of the place recognition module and the map are also shown.

Further Interest in SLAM?

- Tartan SLAM Series:
  - https://theairlab.org/tartanslamseries/
  - https://theairlab.org/tartanslamseries2/

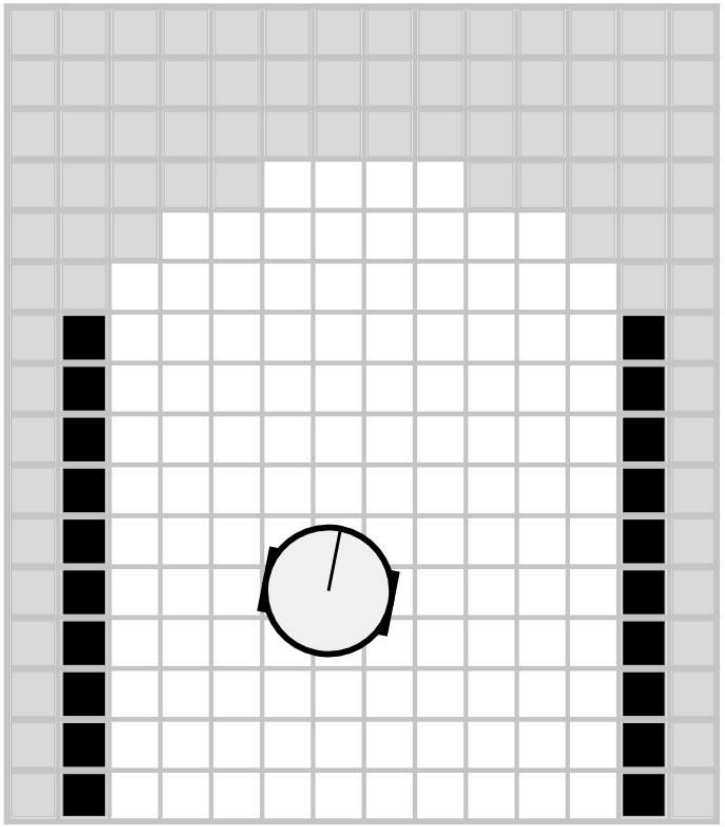
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Visual SLAM

SLAM before Visual SLAM

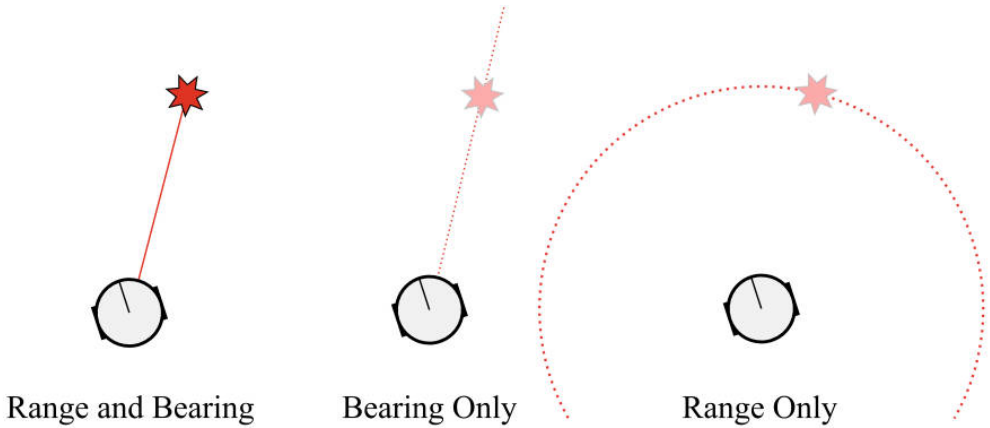
- 2D occupancy maps
- Built from sonar or lidar sensors

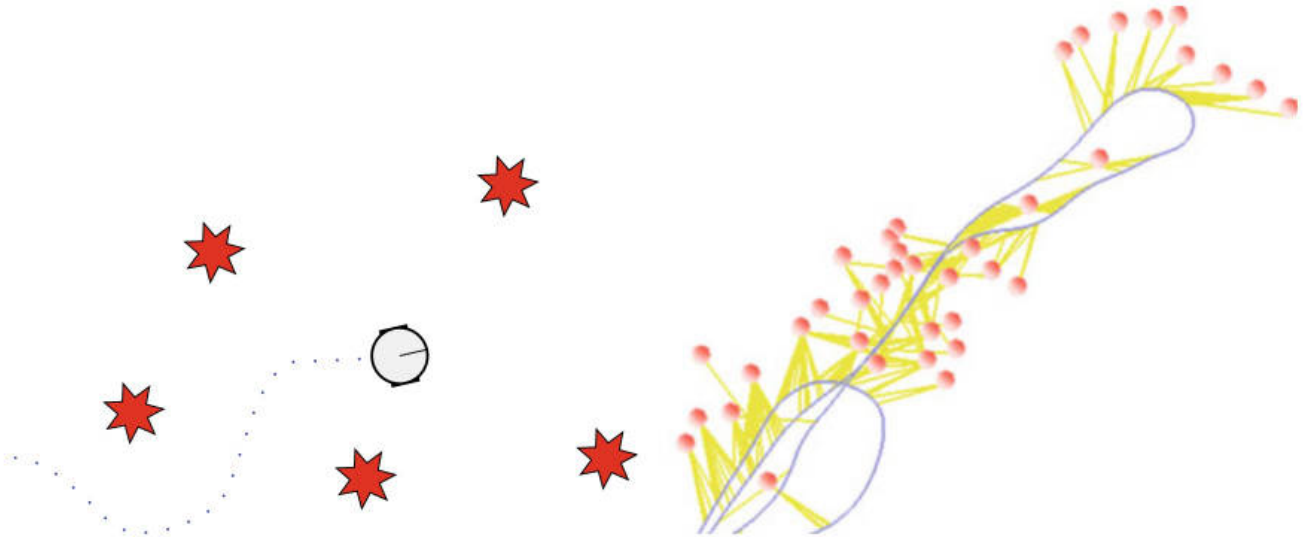




SLAM before Visual SLAM

- Landmark (feature) maps
- Built from sonar or lidar sensors

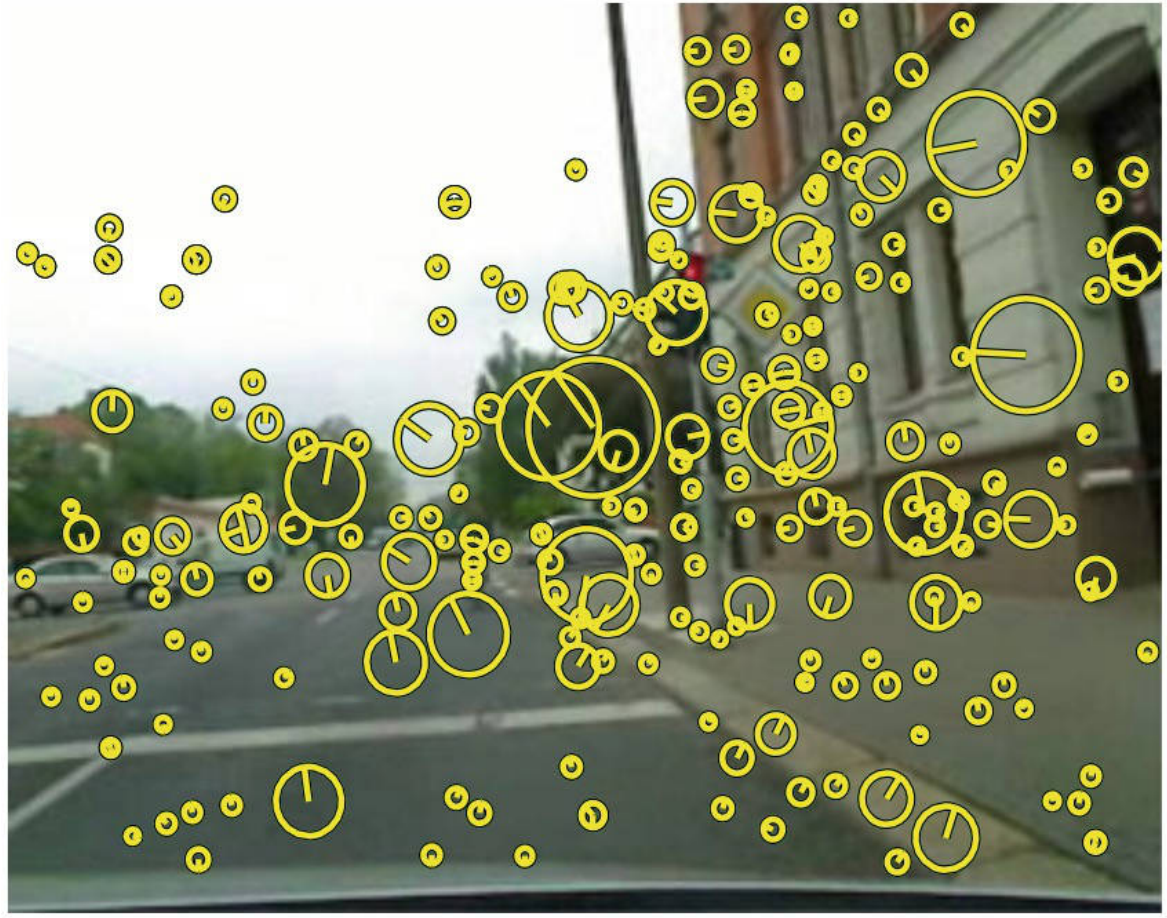




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Why Visual SLAM?

- Cameras provide rich information
  - Geometry
  - Texture
  - Semantics
- Cameras are cheap, compact & lightweight sensors
- Passive sensors



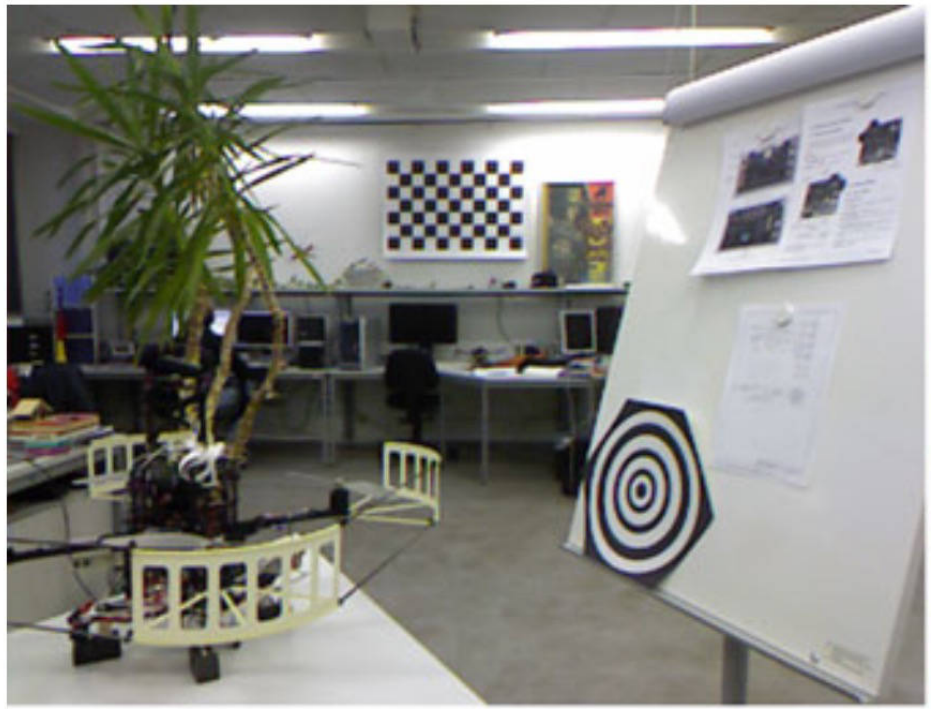
Why Visual SLAM?

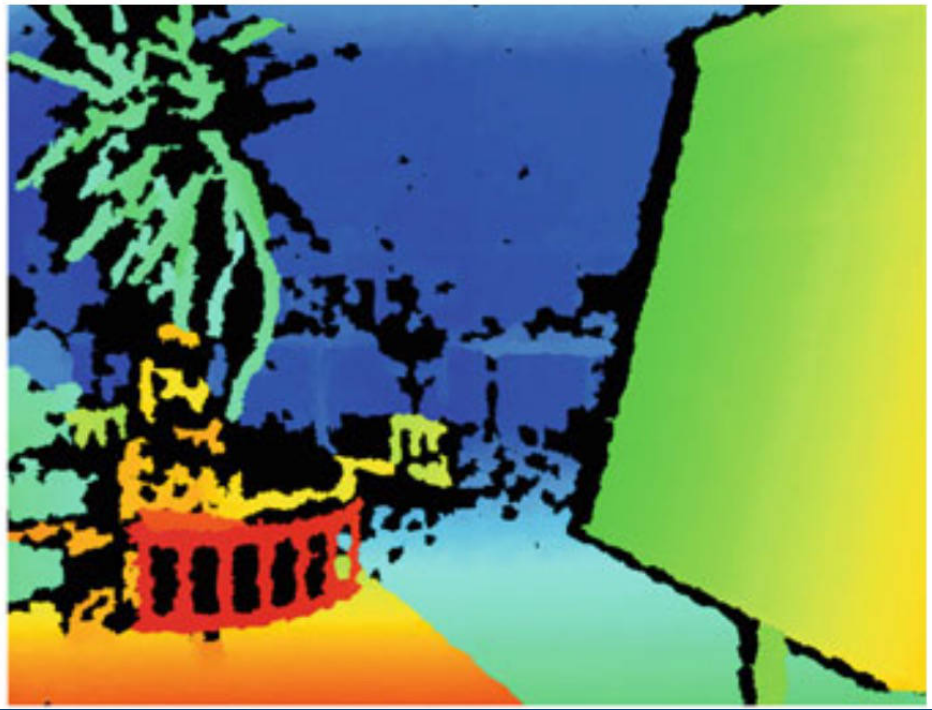




- Advent of Depth Cameras (Kinect) led to even richer perception
- Led to a flood of point cloud & RGB-D SLAM systems



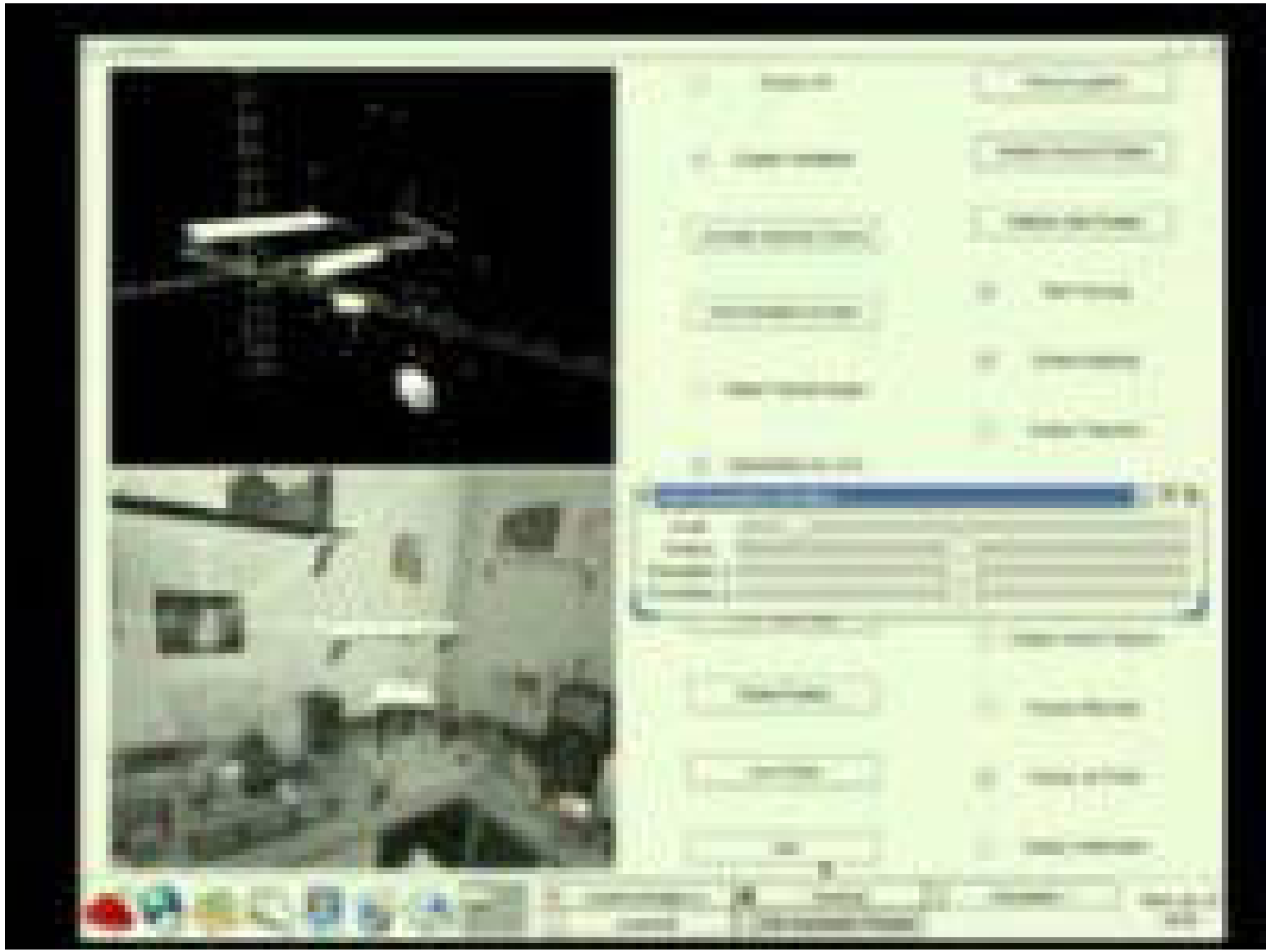




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MonoSLAM

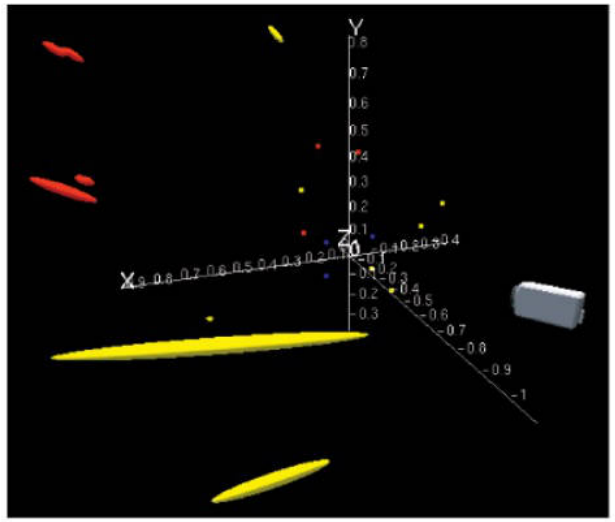
- first real(time) visual SLAM system
- Previous approaches: slow Structure-from-Motion (SfM) based on bundle adjustment

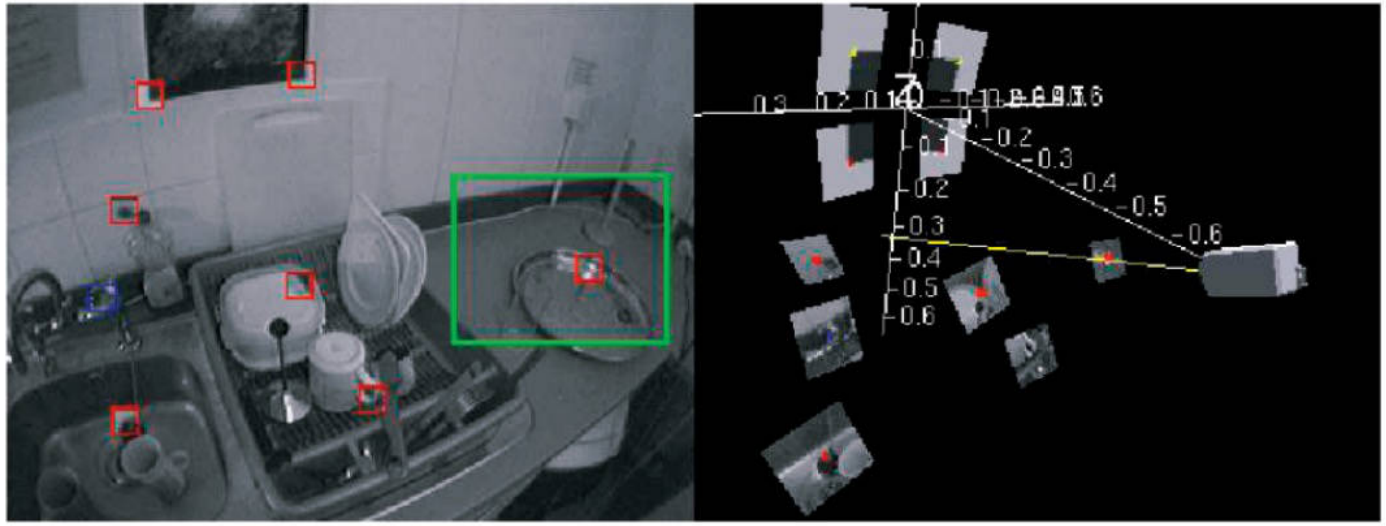


MonoSLAM: Real-Time Single Camera SLAM. Davison et al. TPAMI 2007.

MonoSLAM

MonoSLAM: Real-Time Single Camera SLAM. Davison et al. TPAMI 2007.





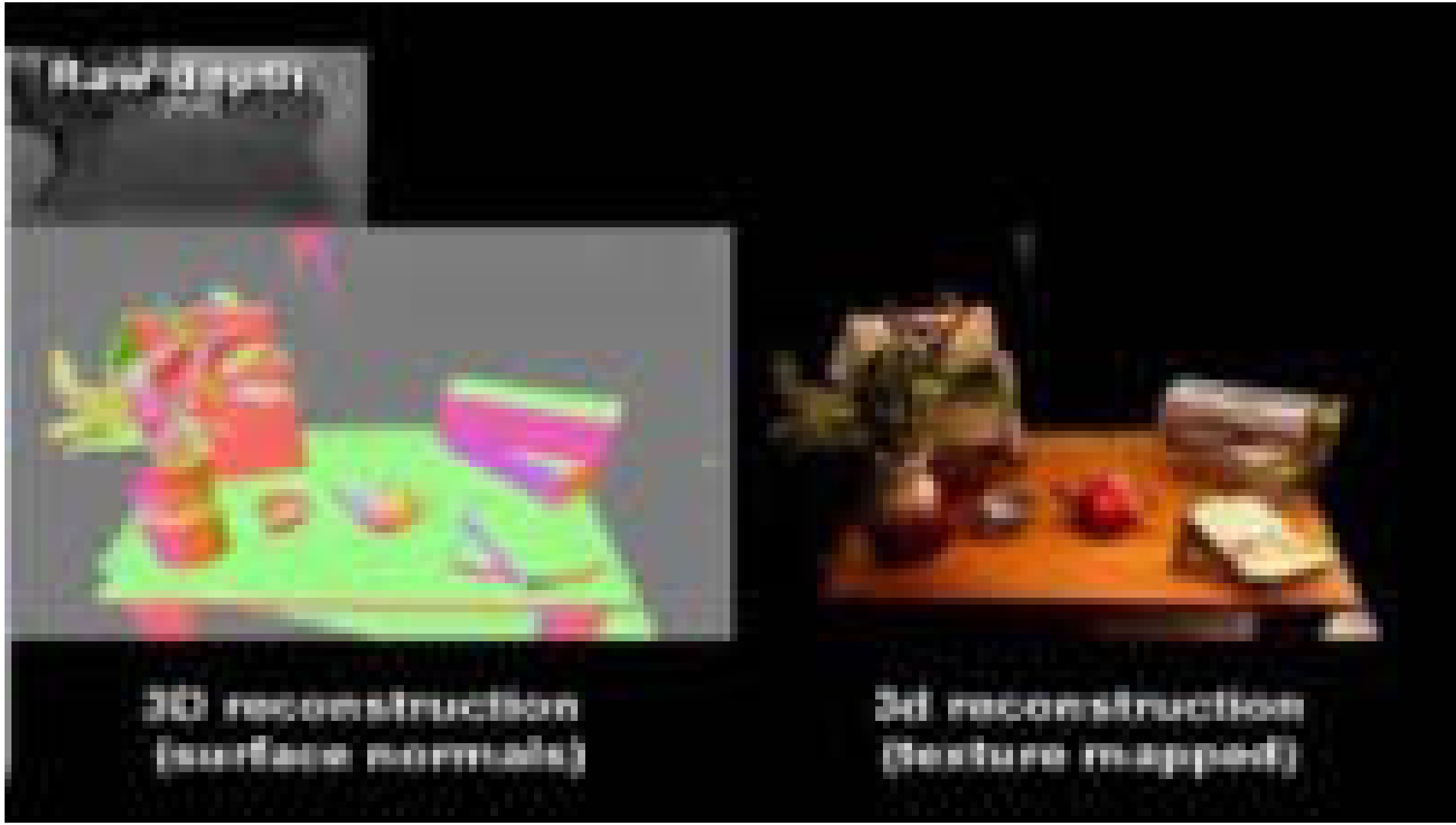
- Pioneering Monocular SLAM (single RGB camera).
- Very sparse map.
- Real-time performance.
- Probabilistic framework (Extended Kalman Filter) to represent and update camera and map state.

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KinectFusion

KinectFusion: Real-time 3D Reconstruction and Interaction Using a Moving Depth Camera. Izadi et al. SIGGRAPH 2011

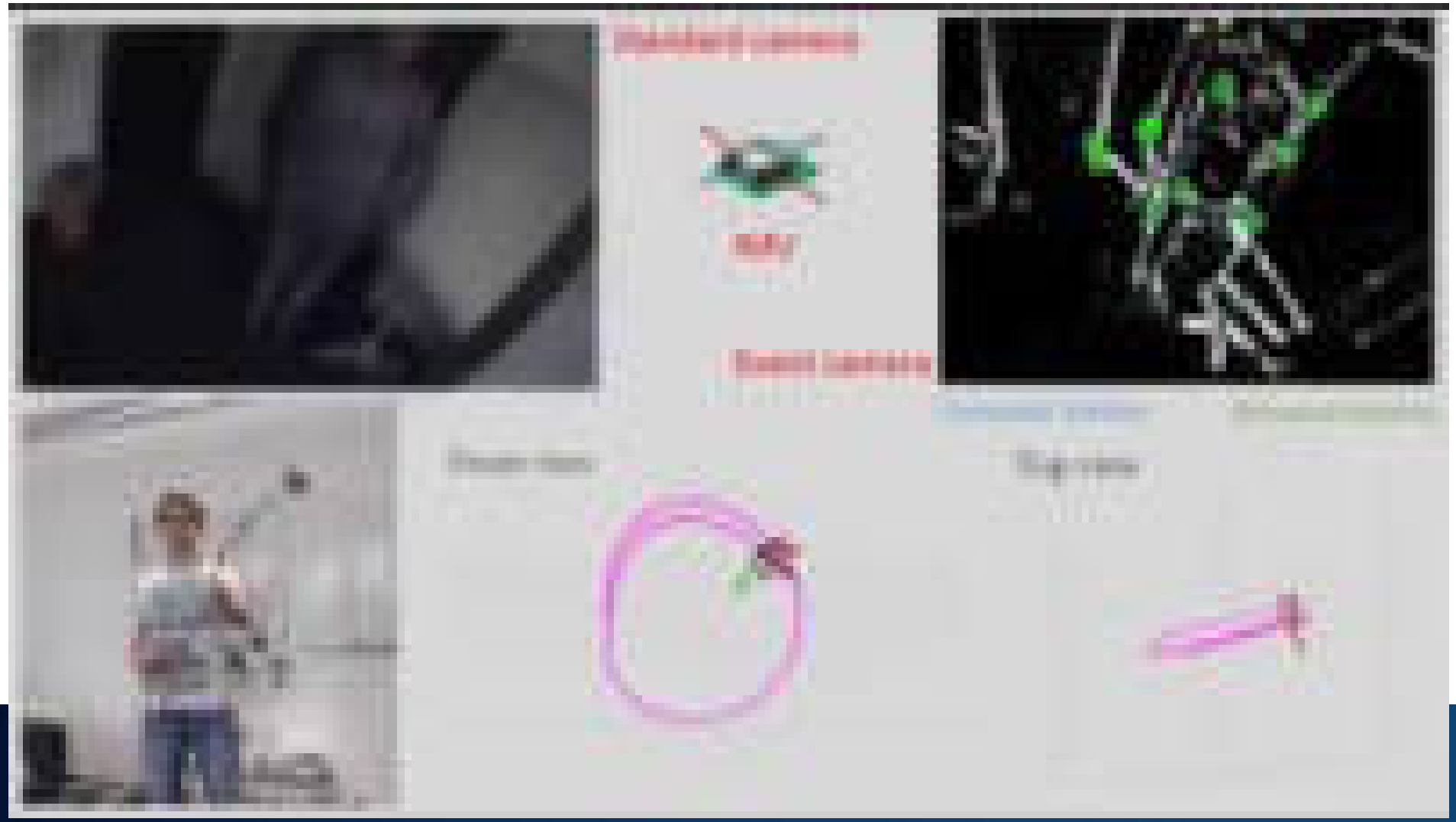
- Pioneered RGB-D SLAM
- Real-time 3D map
- 3D structure represented as Truncated Signed Distance Function (T-SDF)



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Event Camera SLAM

Ultimate SLAM? Combining Events, Images, and IMU for Robust Visual SLAM in HDR and High Speed Scenarios.
Rosinal Vidal et al., RA-L 2018.





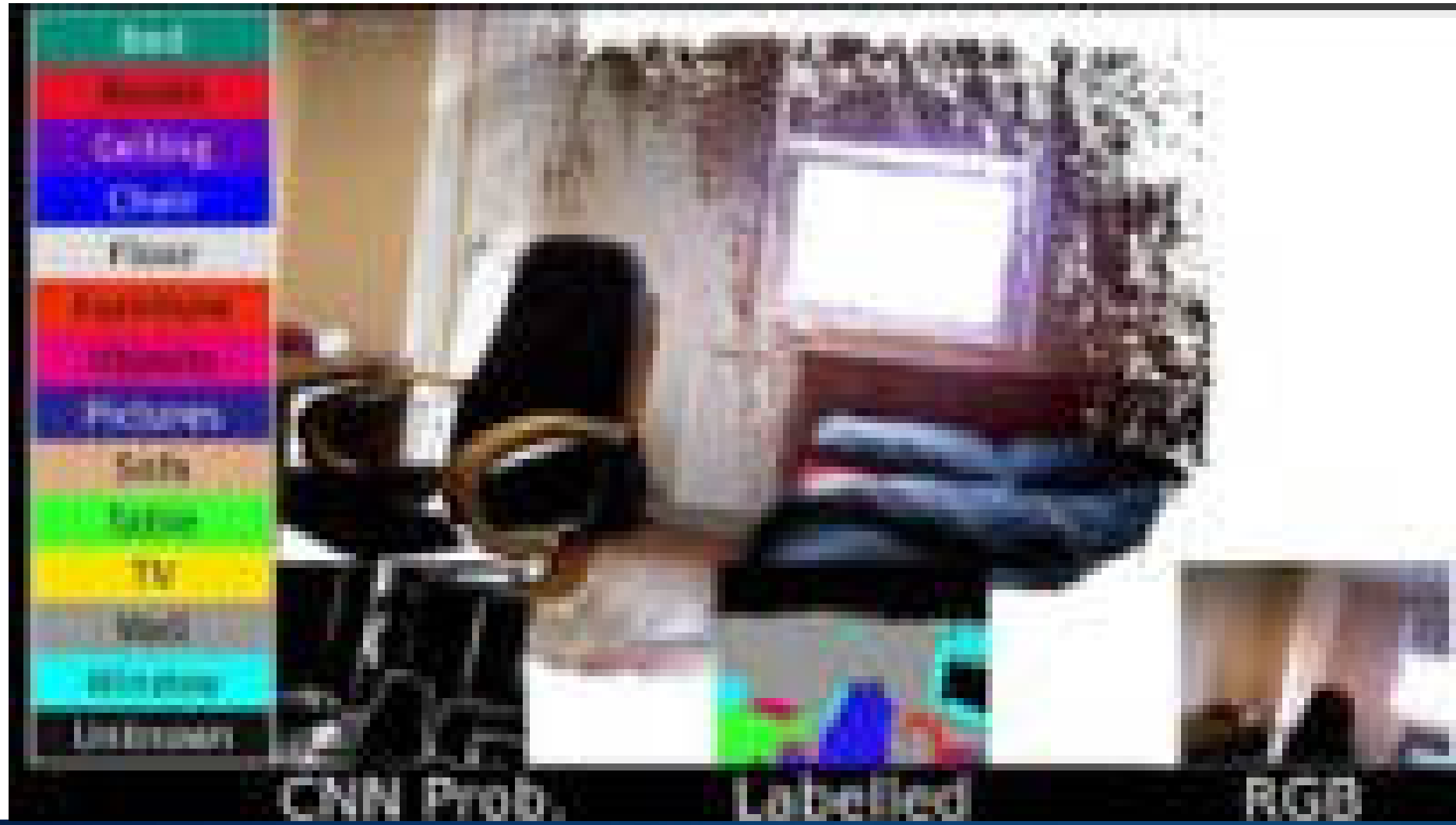
Semantic Visual SLAM

SemanticFusion

SemanticFusion: Dense 3D Semantic Mapping with Convolutional Neural Networks. McCormac et al., ICRA 2017

- Extending dense SLAM to semantic SLAM
- Geometry + semantics

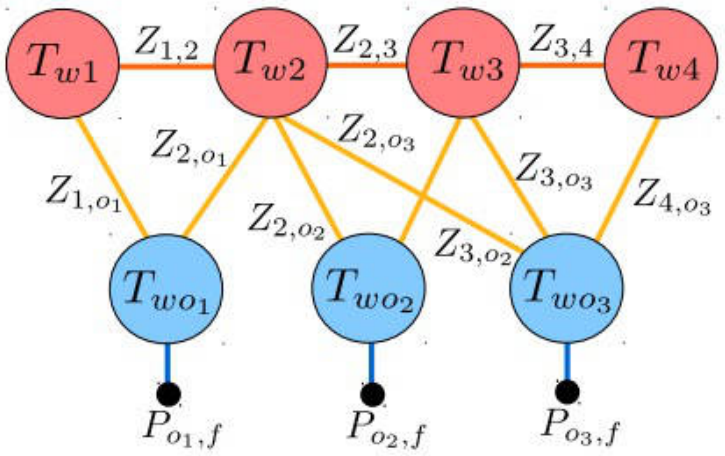
- Use 2D ConvNet for semantics, fuse semantic labels into 3D map



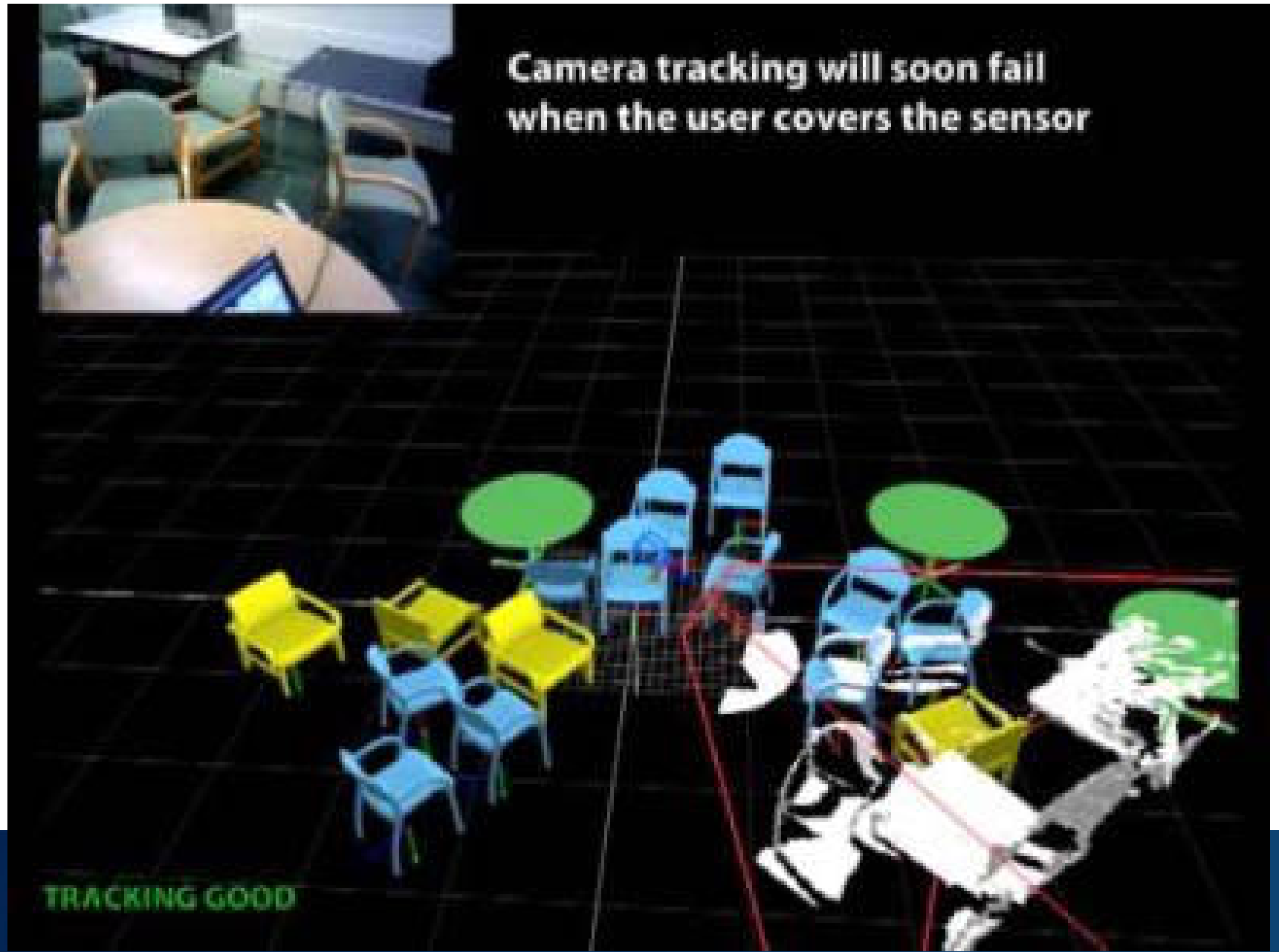
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SLAM++

First object-based SLAM
Objects are landmarks!



SLAM++: Simultaneous Localisation and Mapping at the Level of Objects.
Salas-Moreno et al., CVPR 2013.

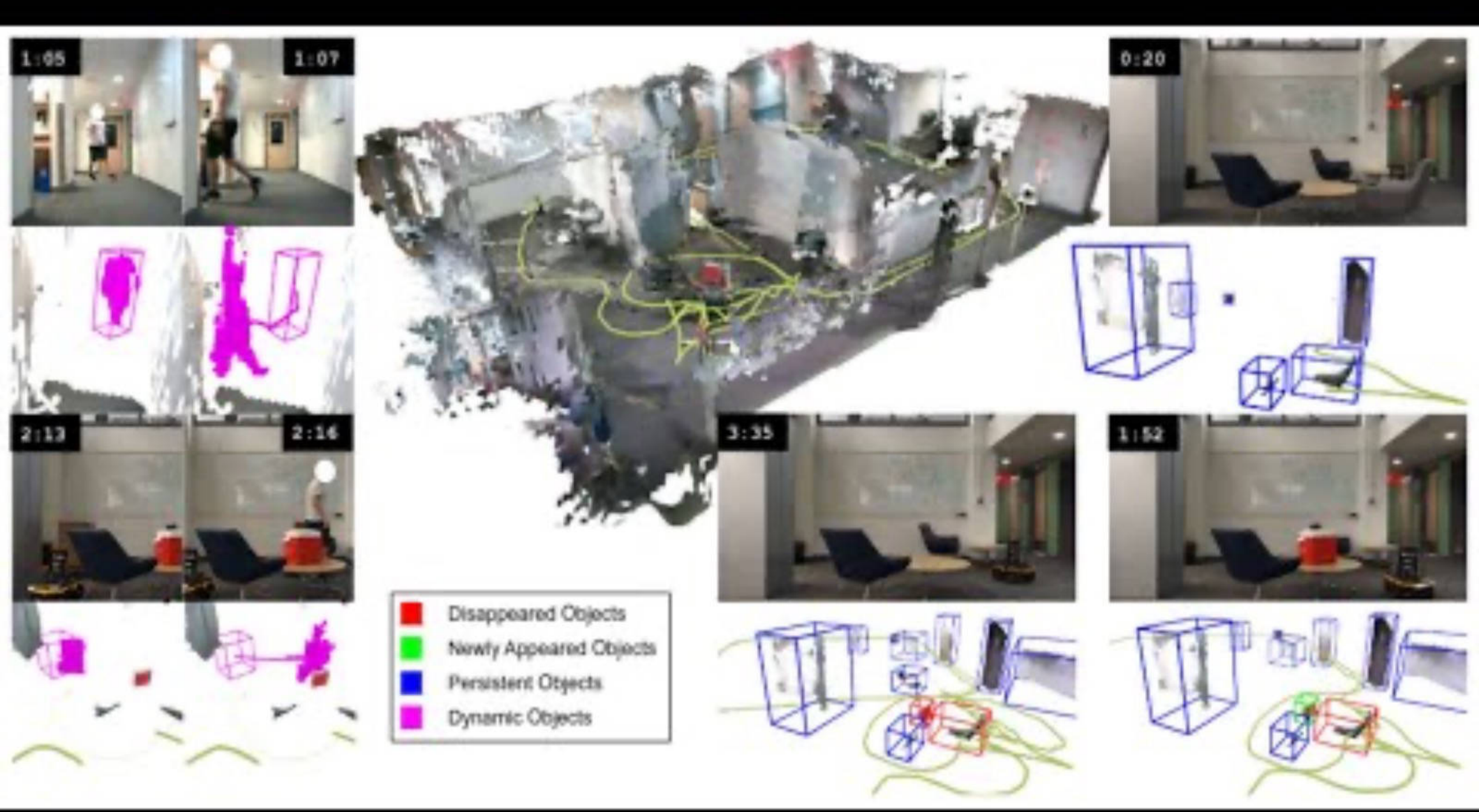


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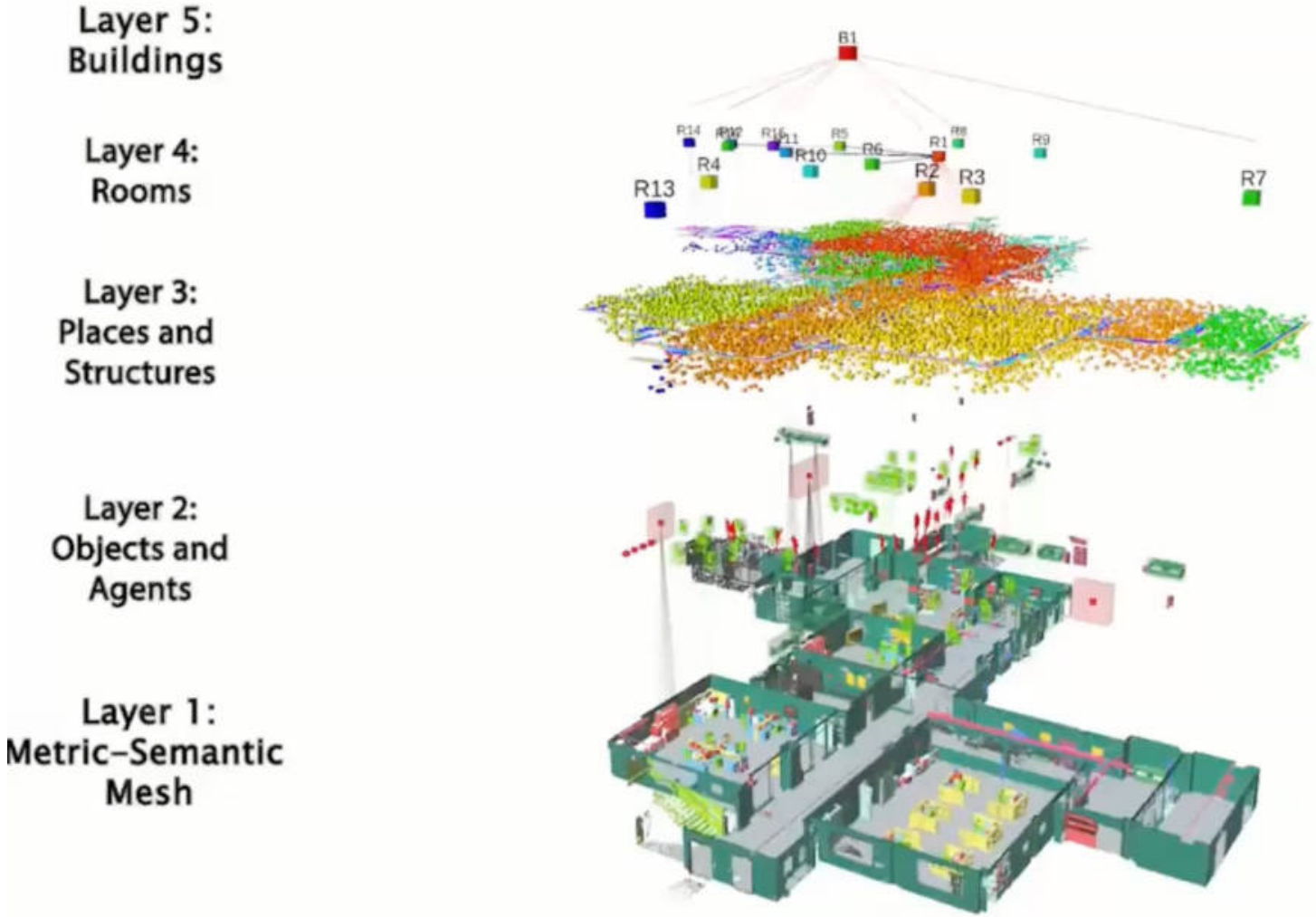
QuadricSLAM

QuadricSLAM: Dual Quadrics from Object Detections as Landmarks in Object-oriented SLAM
Lachlan Nicholson, Michael Milford, Niko Sünderhauf.
IEEE Robotics and Automation Letters, 2018.




1:05
1:07
0:20
2:13
2:16
3:35
1:52
Disappeared Objects
Newly Appeared Objects
Persistent Objects
Dynamic Objects

Scene Graphs



Directed graph, where:

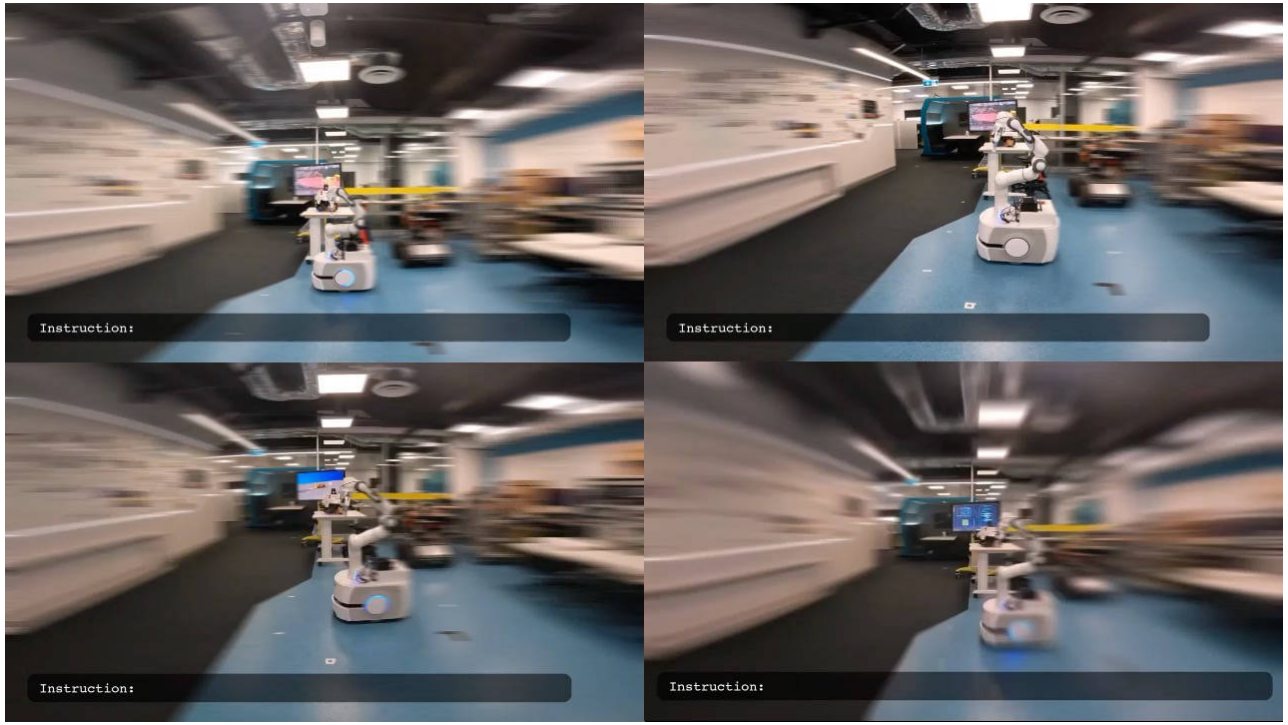
- **nodes** are *spatial concepts* (e.g., concepts grounded in 3D)
- **edges** represent spatio-temporal relations between concepts (e.g., agent "i" in room "j" at time t

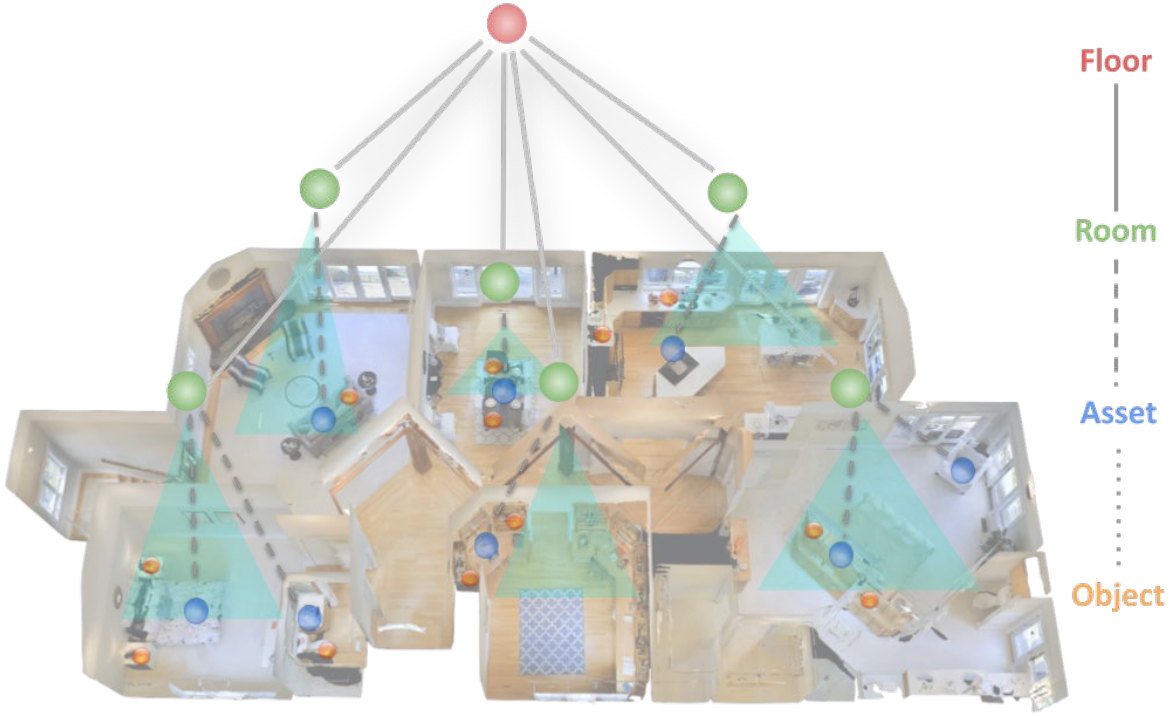
Key insights:

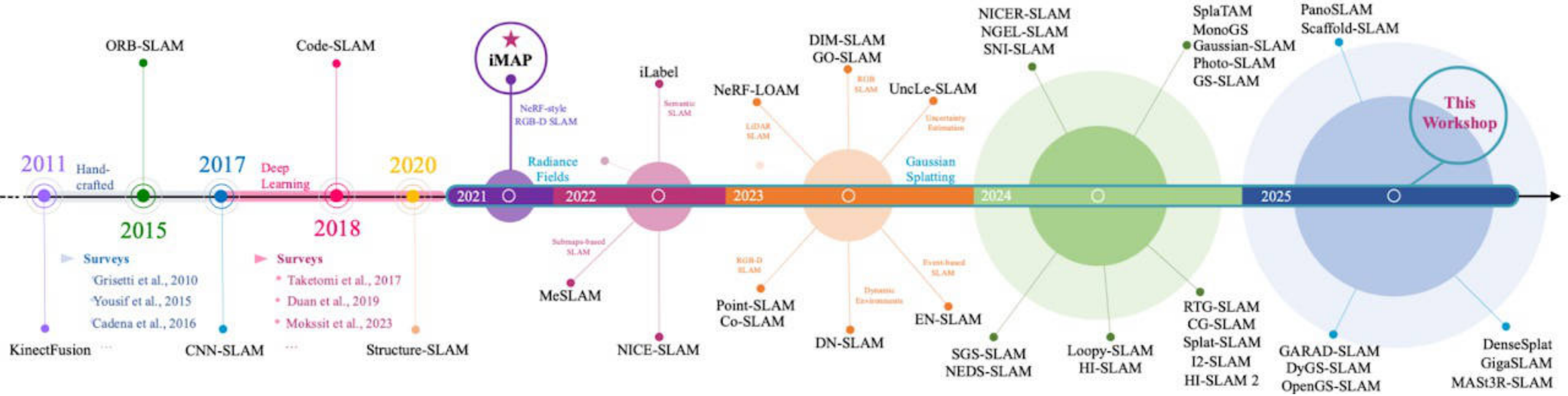
- describe environment at different levels of **abstraction**
- capture **agents** in the scene (robots, humans)
- include **actionable** information for planning and decision-making

[Rosinol, Gupta, Abate, Shi, Carlone. 3D Dynamic Scene Graphs: Actionable Spatial Perception with Places, Objects, and Humans. RSS 2020]
[Armeni et al., 3D scene graph: A structure for unified semantics, 3D space, and camera. ICCV'19]

Large Scale Representation – Scene Graphs







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iMAP

Implicit Mapping and Positioning in Real-Time

ICCV 2021

Edgar Sucar Shikun Liu Joseph Ortiz Andrew Davison

Dyson Robotics Lab, Imperial College London





NICE-SLAM

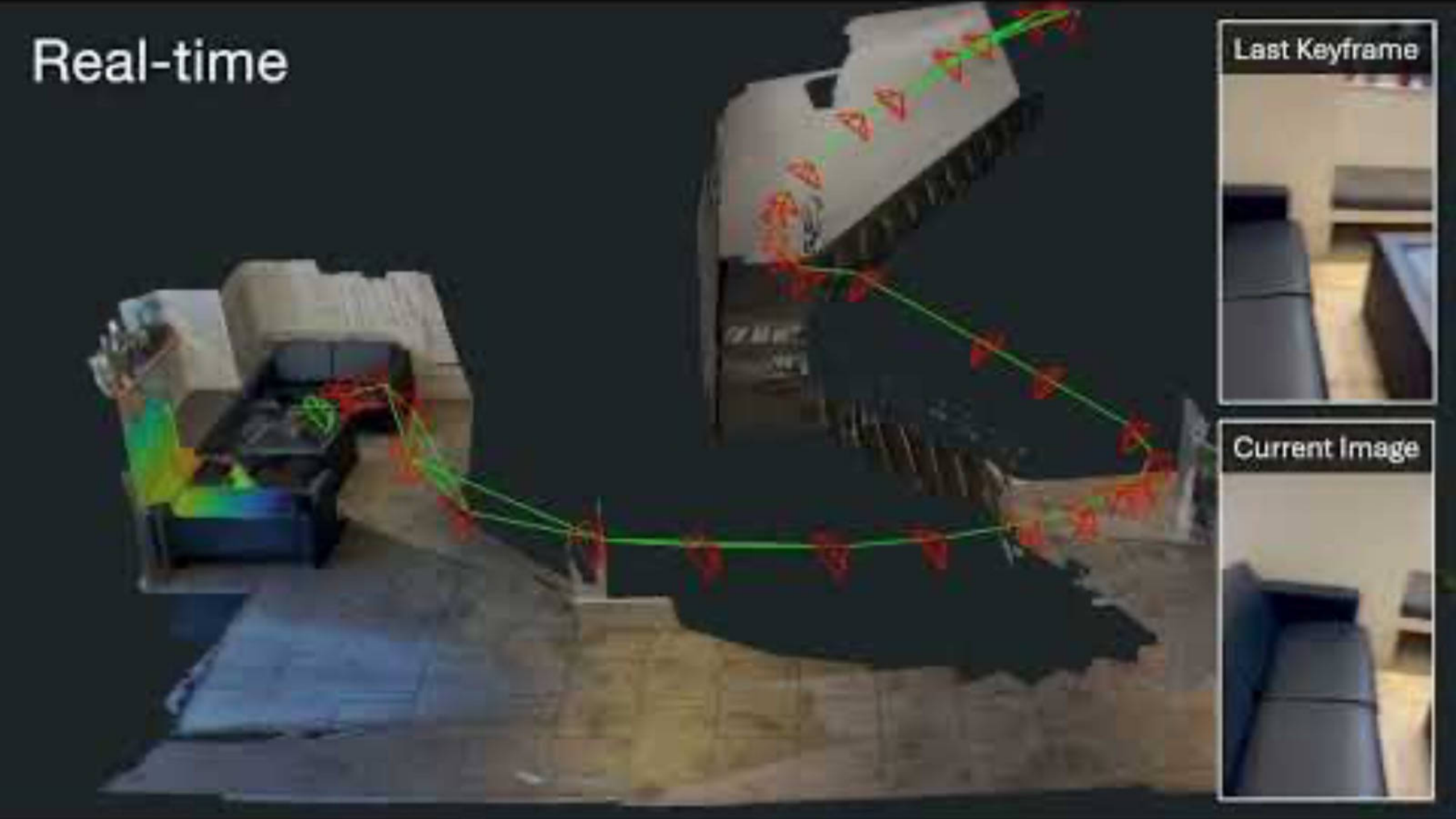
Neural Implicit Scalable Encoding for SLAM

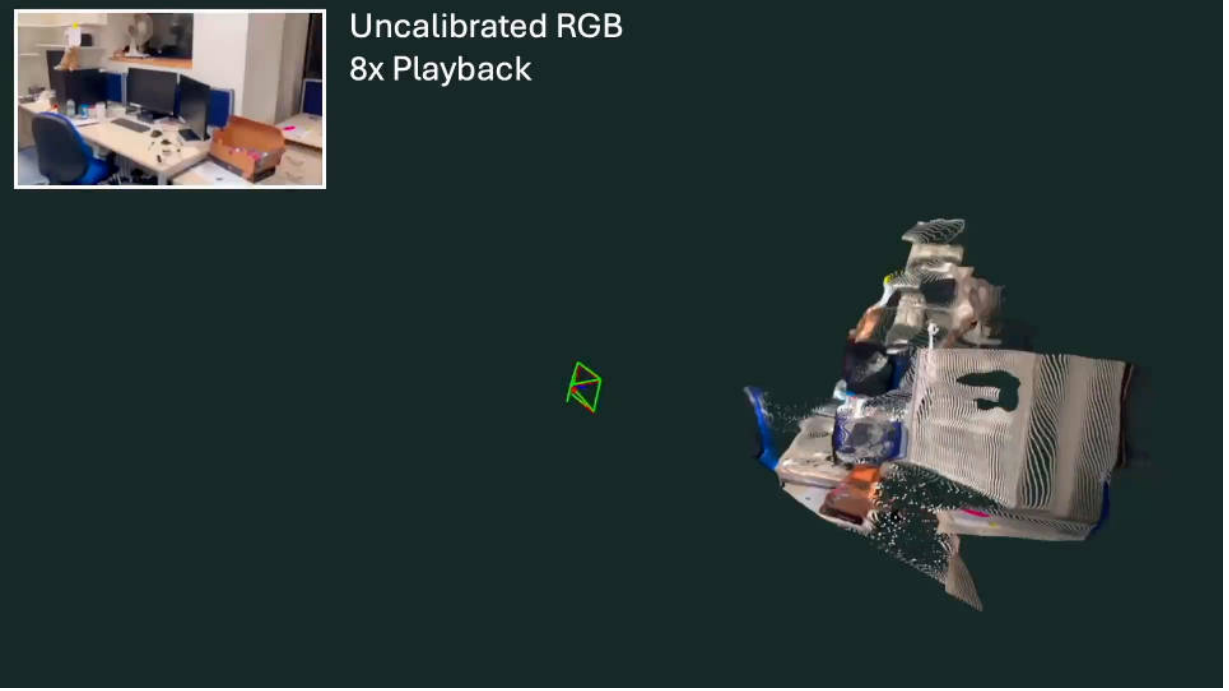
CVPR 2022

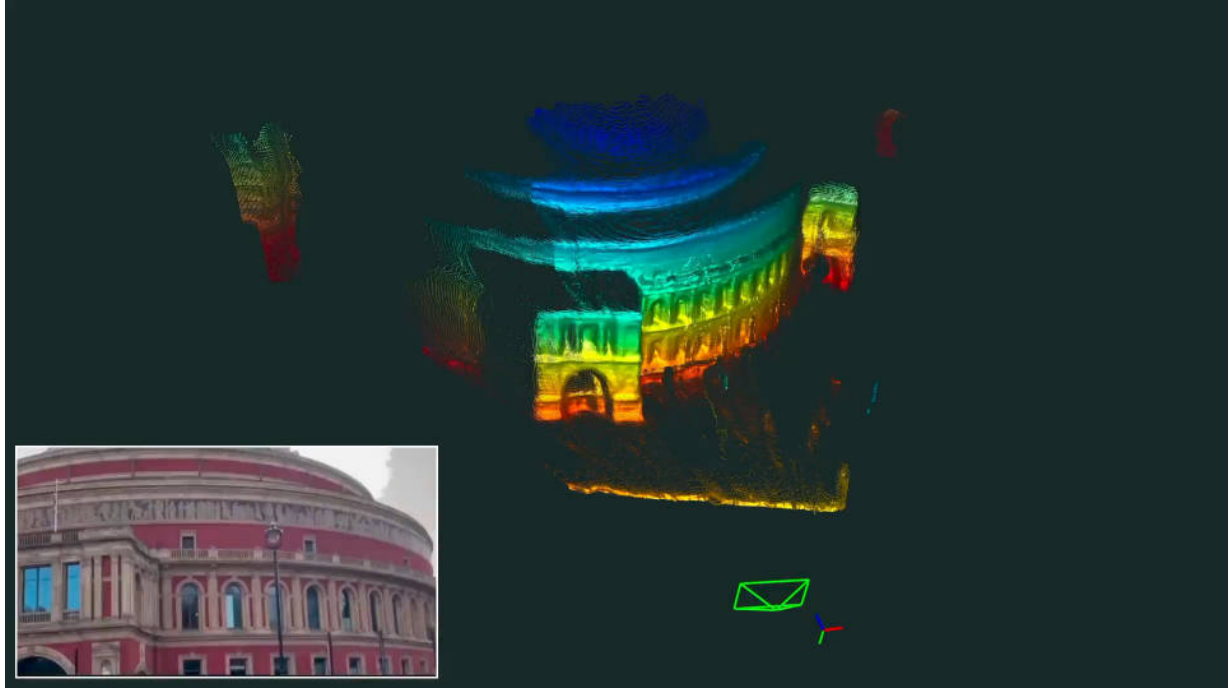
Zihan Zhu* Songyou Peng* Viktor Larsson Weiwei Xu Hujun Bao
Zhaopeng Cui Martin R. Oswald Marc Pollefeys

* Equal Contributions

ETH zürich


Real-time
Last Keyframe
Current Image





New Image

Current Keyframe

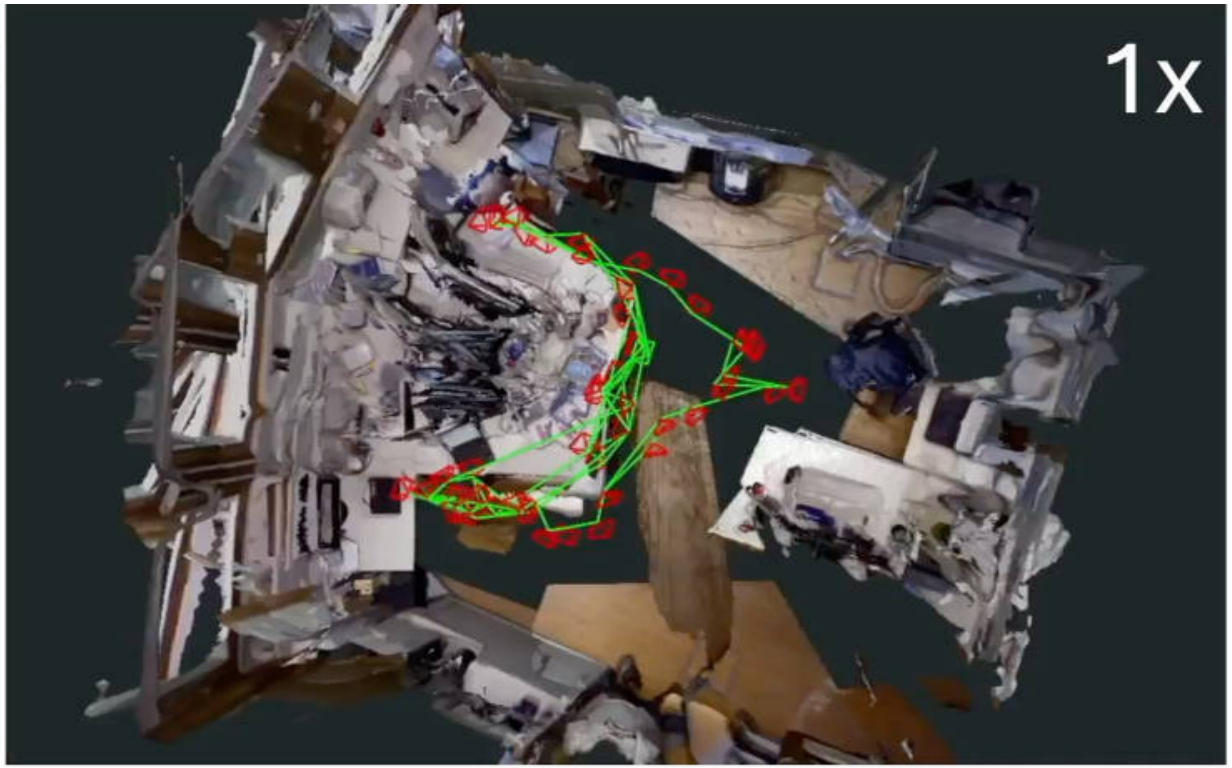
MASt3R (Sec 3.1)

Pointmap Matching (Sec 3.2)

Tracking and Pointmap Fusion (Sec 3.3)

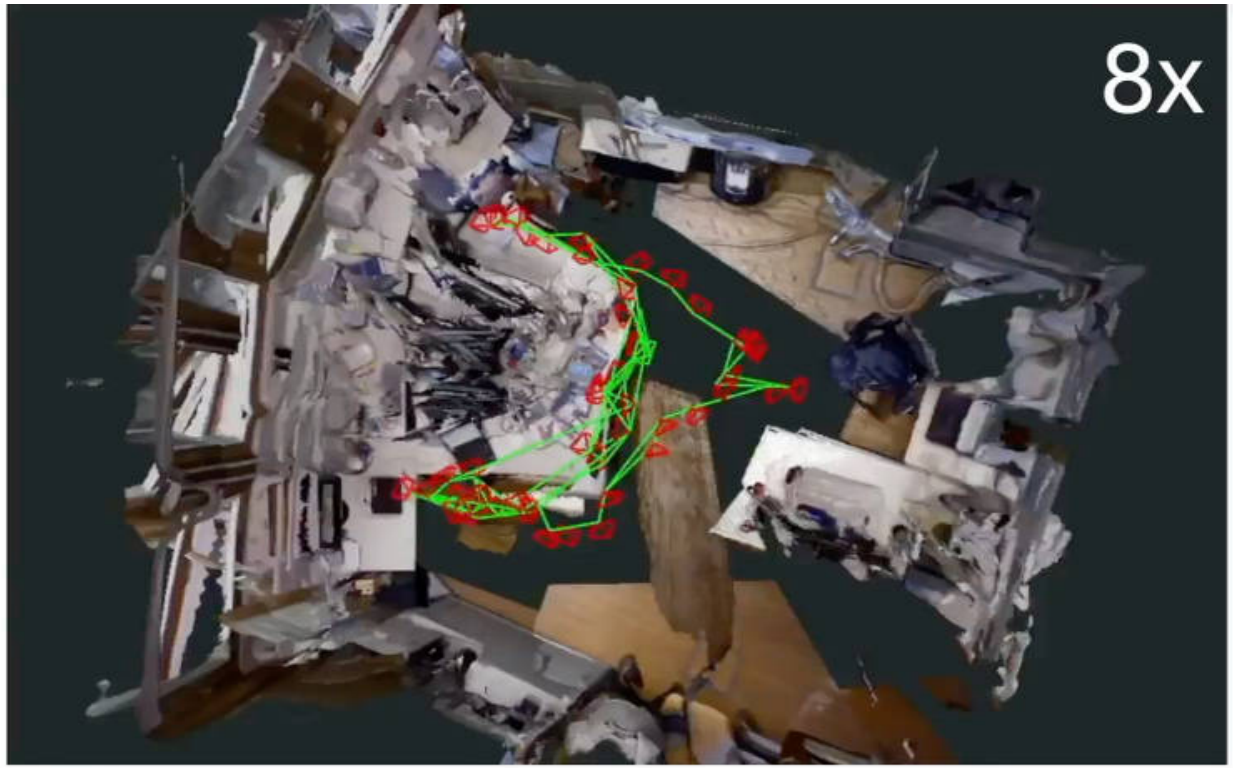
Loop Closure (Sec 3.4) and Global Opt. (Sec 3.5)

Loop Closure Candidates



Gauss-Newton

RMSE ATE 0.118m



Gradient Descent

RMSE ATE 0.147m

We have completed our journey!



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Questions?

Week 8 Onwards

8	Deep Learning Recap – From classification to robotic vision	
9	Learned geometry and modern 3D scene representations	
	Mid-Semester Break	
10	Object detection and open-vocabulary perception	**Assessment 1 due**
11	Segmentation, dense prediction, grasp prediction	
12	Visual representations for retrieval and place recognition	
13	When deep learning for robotic vision fails	